

Full Project Technical Audit

Machine Learning for Cross-Sectional Stock Return Prediction

Complete data, code, modeling, output, consistency, and reproducibility review

Evidence snapshot: 20 July 2026

Repository: `stock_return_ml_project_clean`

This report is documentation only. No existing project file was changed or executed for expensive acquisition or estimation.

Contents

1	Executive Summary	1
1.1	Project at a glance	1
2	Research Objective and Project Overview	1
3	Complete Chronological Project Story	2
3.1	Stage-by-stage narrative	2
4	Repository Architecture	4
5	Complete File Inventory	4
6	File-by-File Technical Documentation	4
7	Data Sources and Data Lineage	4
7.1	End-to-end flowchart	4
8	Data Construction and Timing Decisions	4
9	Final Dataset Audit	4
9.1	Major dataset validation table	5
10	Modeling Framework	7
11	Model-by-Model Documentation	7
12	Validation and Test Design	7
12.1	Split counts	7
13	Results	7
13.1	Frozen final-test comparison	7
14	Interpretation	9
15	What Was Done Well	9
16	Limitations and Questionable Choices	9
17	Leakage and Look-Ahead-Bias Audit	10
17.1	Verification matrix	10
18	Reproducibility Audit	13
19	Consistency and Staleness Audit	13
19.1	Evidence-supported inconsistency register	13
20	Required Checks Before Submission	16
21	Recommended Future Improvements	16
22	Final Overall Assessment	16
22.1	Scorecard	16
23	Reproduction Guide	17
23.1	Required environment	17
23.2	Three run levels	17

24 Glossary	18
A Full Directory Tree	18
B Complete File Inventory	23
B.1 Identity and classification inventory	23
B.2 Producer, consumer, and reproducibility inventory	48
C Complete File-by-File and Function Documentation	70
C.1 File 1: .gitignore	70
C.2 File 2: README.md	70
C.3 File 3: audit report.txt	70
C.4 File 4: existing output files.txt	70
C.5 File 5: original directory tree.txt	71
C.6 File 6: 01 fixed linear models.py	71
C.7 File 7: 02 fixed tree models.py	73
C.8 File 8: 03 compare fixed models.py	75
C.9 File 9: 01 optimized linear models.py	77
C.10 File 10: 02 optimized tree models.py	79
C.11 File 11: 03 compare optimized models.py	81
C.12 File 12: 04 compare fixed vs optimized.py	83
C.13 File 13: estimation.py	85
C.14 File 14: evaluation.py	87
C.15 File 15: gradient boosting inventory.txt	89
C.16 File 16: README.md	89
C.17 File 17: README MODEL WORKFLOW.md	89
C.18 File 18: 02 fixed tree models.py	89
C.19 File 19: 03 compare fixed models.py	92
C.20 File 20: 02 tune tree models.py	94
C.21 File 21: 03 compare tuning results.py	96
C.22 File 22: 02 optimized tree models.py	98
C.23 File 23: 03 compare optimized models.py	100
C.24 File 24: 04 compare fixed vs optimized.py	102
C.25 File 25: 01 time series robustness.py	104
C.26 File 26: 01 final test evaluation.py	106
C.27 File 27: 01 portfolio sorts.py	108
C.28 File 28: 02 compare portfolio results.py	110
C.29 File 29: 03 final model and portfolio summary.py	112
C.30 File 30: Data Construction Guide.tex	114
C.31 File 31: README.md	114
C.32 File 32: decision tree rules.txt	114
C.33 File 33: diagnostic warnings.txt	114
C.34 File 34: top elastic net coefficients skipped.txt	115
C.35 File 35: report notes.md	115
C.36 File 36: requirements.txt	115
C.37 File 37: run pipeline.sh	115
C.38 File 38: init .py	116
C.39 File 39: 01 create locked stock universe.py	116
C.40 File 40: 02 create fama french rf input.py	118
C.41 File 41: 03 download market inputs.py	120
C.42 File 42: 04 create welch goyal input.py	122
C.43 File 43: init .py	124
C.44 File 44: manifest.py	124
C.45 File 45: config.py	126

C.46 File 46: 01 build clean yahoo daily.py	126
C.47 File 47: 02 build monthly stock features.py	128
C.48 File 48: 03 add fundamentals and macro.py	130
C.49 File 49: 04 build raw kelly dataset.py	132
C.50 File 50: 05 clean and rank normalize.py	134
C.51 File 51: init .py	136
C.52 File 52: Pipeline Overview.tex	136
C.53 File 53: feature definitions.py	136
C.54 File 54: README.md	136
C.55 File 55: README MODEL WORKFLOW.md	137
C.56 File 56: init .py	137
C.57 File 57: README.md	137
C.58 File 58: README MODEL WORKFLOW.md	137
C.59 File 59: 02 fixed tree models.py	138
C.60 File 60: 02 tune tree models.py	140
C.61 File 61: 03 compare tuning results.py	142
C.62 File 62: 02 optimized tree models.py	144
C.63 File 63: 01 time series robustness.py	146
C.64 File 64: 01 final test evaluation.py	148
C.65 File 65: 01 fixed linear models.py	150
C.66 File 66: 02 fixed tree models.py	152
C.67 File 67: 03 compare fixed models.py	154
C.68 File 68: 04 diagnose fixed models.py	156
C.69 File 69: init .py	158
C.70 File 70: 01 tune linear models.py	158
C.71 File 71: 02 tune tree models.py	160
C.72 File 72: 03 compare tuning results.py	162
C.73 File 73: init .py	164
C.74 File 74: 01 optimized linear models.py	164
C.75 File 75: 02 optimized tree models.py	166
C.76 File 76: 03 compare optimized models.py	168
C.77 File 77: 04 compare fixed vs optimized.py	170
C.78 File 78: init .py	172
C.79 File 79: 01 time series robustness.py	172
C.80 File 80: init .py	174
C.81 File 81: 01 final test evaluation.py	174
C.82 File 82: init .py	176
C.83 File 83: 00 utils.py	176
C.84 File 84: 01 model interpretation.py	178
C.85 File 85: 02 feature importance.py	180
C.86 File 86: 03 create result tables.py	182
C.87 File 87: 04 create figures.py	184
C.88 File 88: 05 report notes.py	186
C.89 File 89: README.md	188
C.90 File 90: init .py	188
C.91 File 91: run all.py	188
C.92 File 92: init .py	190
C.93 File 93: data.py	190
C.94 File 94: estimation.py	192
C.95 File 95: evaluation.py	194
C.96 File 96: Makefile	196
C.97 File 97: README.md	196
C.98 File 98: paper.tex	196

C.99 File 99: references.bib	196
D Complete Function Inventory	197
E Dataset Schemas	208
F Critical Decision Register	208
G Risk Register	211
H Hard-Coded Parameter and Seed Inventory	214
I Output Lineage	214
J Unused, Duplicate, Orphan, or Obsolete Inventory	214
K Final Submission Checklist	215
L Assumption and Readiness Register	217

1 Executive Summary

Audit convention. Statements marked **Verified fact** were checked directly against current code or outputs. **Interpretation** explains implications. **Potential problem** identifies evidence-supported risk. **Recommendation** proposes a future check but does not alter the project. Where evidence is absent, this report states: **Not verifiable from the current repository.**

The project predicts next-month stock excess returns in a monthly ticker panel using market, technical, accounting, industry, and macroeconomic information. The final model-ready file contains 212,794 observations, 663 tickers, and 499 predictors from January 1990 through December 2025. The development period ends in December 2019 and the frozen final test covers January 2020 through December 2025. Six final specifications are compared: a historical mean, OLS-3, PLS, Elastic Net, Decision Tree, and Random Forest.

Verified finding. PLS ranks first on final monthly MSE, but its OOS R-squared is only 0.00179. Random Forest and OLS-3 also achieve small positive OOS R-squared values. Elastic Net collapses to zero coefficients, and the single Decision Tree materially underperforms the historical mean. These results support weak statistical predictability, not strong economic profitability.

Most urgent checks. Enforce next-calendar-month target continuity, resolve macro publication lags, repair the stale shell pipeline, reconcile tuning documentation with actual 1990-2014 tuning and minimum-MSE selection, persist failed Yahoo tickers, and lock software versions.

1.1 Project at a glance

Item	Verified repository value
Research question	Can stock, market, accounting, industry, and macro predictors forecast next-month stock excess returns out of sample?
Unit	Ticker-month
Target	target_excess_return_next_1m
Coverage	1990-01-31 to 2025-12-31
Universe	Locked 867-ticker current-plus-historical Wikipedia union; 663 in final panel
Main sources	Yahoo Finance, WRDS/Compustat, Ken French RF, cleaned Welch-Goyal input, Wikipedia
Predictors	47 ranked characteristics, 4 market/VIX, 8 macro, 64 SIC2 dummies, 376 interactions
Development/test	1990-2019 / 2020-2025
Actual tuning folds	Annual expanding folds within 1990-2014, validation years 2005-2014
Primary metric	Equal-weighted monthly MSE
Best final model	PLS: monthly MSE 0.0137866, OOS R2 0.0017914
Main strength	Strong chronological separation and verified six-month backward Compustat timing
Main limitation	Weak gains plus timing/continuity and documentation inconsistencies

2 Research Objective and Project Overview

The active repository implements a regression problem, not classification. Predictors dated at month t are intended to forecast stock return in $t+1$ less the $t+1$ risk-free rate. The design follows the empirical asset-pricing machine-learning tradition: combine many firm characteristics with macro states, compare flexible methods against transparent benchmarks, and reserve a chronologically later test period.

3 Complete Chronological Project Story

The workflow begins with one-time local inputs, then downloads individual stock histories, cleans daily observations, aggregates month-end features, constructs the target, attaches conservatively lagged accounting statements, merges macro states and industries, performs cross-sectional imputation/ranking, estimates fixed and tuned models, runs robustness checks, opens a final test once, and creates frozen interpretation and paper artifacts. The current shell wrapper no longer represents the actual model-stage paths.

3.1 Stage-by-stage narrative

Step	Action	Script	Inputs	Outputs	Verification
1	Create permanent inputs	src/acquisition/01-04	Wikipedia, Ken French, Yahoo market series, external Welch-Goyal, Compustat supplied separately	input/*	Check manifest and coverage
2	Build daily stock file	src/data/01_build_clean_yahoo_daily.py	locked universe	raw/clean daily CSV and quality reports	Check row/ticker counts and removals
3	Build monthly stock panel	src/data/02_build_monthly_stock_features.py	clean daily, GSPC, VIX, RF	monthly panel and target	Check month-end, duplicates, continuity
4	Add fundamentals	src/data/03_add_fundamentals_and_macro.py	monthly panel, raw Compustat	clean Compustat and merged panel	Check available__month <= month
5	Build raw Kelly panel	src/data/04_build_raw_kelly_dataset.py	fundamental panel, Welch-Goyal	123-predictor raw panel	Check macro lag
6	Clean and rank	src/data/05_clean_and_rank_normalize.py	raw Kelly panel	499-predictor parquet	Check missing, finite, duplicate, rank ranges
7	Fixed models	src/models/step_01_fixed/*	final parquet	fixed metrics/predictions	No test rows
8	Tune	src/models/step_02_tuning/01-03	train through 2014	grids and best parameters	Annual chronological folds
9	Validate optimized	src/models/step_03_optimization/*	train/validation	optimized comparisons	No test rows
10	Robustness	src/models/step_04_robustness/*	train/validation	expanding vs rolling metrics	Check 2015-2019 only
11	Final test	src/models/step_05_test/*	development and saved parameters	test predictions/metrics	One-time process not historically provable
12	Interpret/report	src/models/step_06_interpretation/*andthesis/paper.tex	frozen outputs	tables, figures, notes, paper	No model refit

4 Repository Architecture

The pre-audit snapshot contains 373 files outside .git and the new audit folder. Active logic lives under src; permanent inputs under input; frozen data and model results under output; archival code/results under backup and legacy; and the current course paper under thesis.

5 Complete File Inventory

Every file is listed in Appendix B with type, category, role, producer, consumer, essential status, and reproducibility assessment. Cache files, OS metadata, build auxiliaries, backups, and legacy Gradient Boosting artifacts are deliberately included rather than silently omitted.

6 File-by-File Technical Documentation

Appendix C gives a dedicated subsection for each Python, shell, LaTeX, Markdown, bibliography, text, Makefile, dependency, and ignore file. Active files receive decision and quality notes; archival files are explicitly separated from current evidence.

7 Data Sources and Data Lineage

The permanent-input design successfully prevents small online datasets from changing during a normal build. Individual stock prices remain an online acquisition inside data file 01. Raw Compustat extraction credentials and query are not stored; therefore exact reconstruction of that input is not verifiable from the current repository. No active FRED ingestion exists.

7.1 End-to-end flowchart

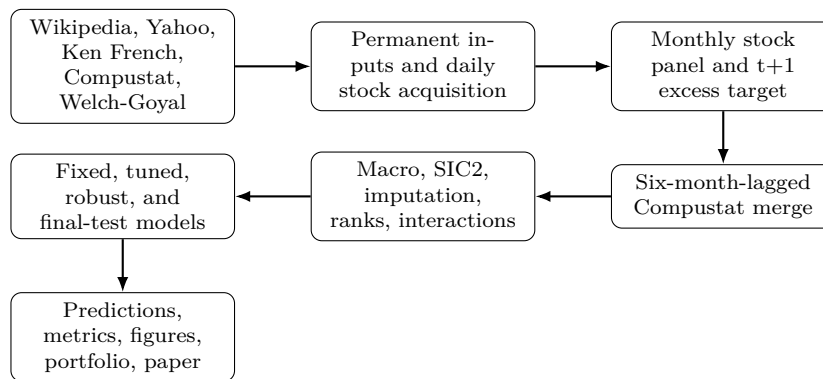


Figure 1: Verified high-level data and output lineage.

8 Data Construction and Timing Decisions

Daily adjusted prices produce simple returns and monthly technical characteristics. Current month-end information predicts the subsequent target. Momentum explicitly lags current return. Compustat is delayed six months and merged backward. Same-month Welch-Goyal values are merged without the intended one-month shift, and target shifting does not enforce calendar continuity.

9 Final Dataset Audit

The final parquet is sorted by ticker and month, has unique column names, zero missing values, zero infinities, and zero duplicate ticker-month rows. Its 499 predictors exactly match the predictor inventory.

Near-duplicate predictors were not exhaustively tested because a complete 499-by-499 dependence audit is computationally substantial; this is not verifiable from the current repository outputs.

9.1 Major dataset validation table

Dataset	Path	Shape	Coverage	Entities	Missing	Duplicates	Notes
Locked universe	input/stock_universe_locked.csv	867 x 1	n/a	867	0	0	Permanent Wikipedia-derived current-plus-historical union
Fama-French RF	input/fama_french_rf_monthly.csv	1,199 x 2	1926-07 to 2026-05	n/a	0	0 months	RF decimal, month-end
GSPC daily	input/market_gspc_daily.csv	9,845 x 7	1987-01-02 to 2026-01-30	1 series	0	0 dates	auto_adjust=False
VIX daily	input/market_vix_daily.csv	9,087 x 7	1990-01-02 to 2026-01-30	1 series	0	0 dates	auto_adjust=False
Raw Compustat	input/raw/compustat_annual_1980_2025.csv	21,575 x 84	1980-01-31 to 2026-01-31	649 tickers	148,161	0 ticker-date	Unsorted input; annual statements
Welch-Goyal input	input/external/welch_goyal_macro_1990_2025.csv	432 x 10	1990-01 to 2025-12	n/a	0	0 months	Nine values plus month; only eight are used
Clean daily prices	output/data/final/daily_prices_clean_1987_2026.csv	4,701,144 x 8	1987-01-02 to 2026-01-30	670	0	0 ticker-date	Sorted
Monthly stock panel	output/data/final/monthly_stock_panel_with_targets_1990_2025.csv	212,794 x 30	1990-01 to 2025-12	663	43,229	0 ticker-month	54 infinities before final cleaning
Fundamental panel	output/data/intermediate/monthly_panel_with_compustat_macro_1990_2025.csv	212,794 x 61	1990-01 to 2025-12	663	497,813	0 ticker-month	207,739 matched rows; no lag violation
Raw Kelly panel	output/data/final/model_dataset_kelly_raw_full_1990_2025.csv	212,794 x 126	1990-01 to 2025-12	663	482,648	0 ticker-month	123 base predictors
Ranked model panel	output/data/final/model_dataset_kelly_ranked_full_1990_2025.parquet	212,794 x 502	1990-01 to 2025-12	663	0	0 ticker-month	499 predictors; no infinities
Final test predictions	output/models/test/final_test_predictions.parquet	268,968 x 6	2020-01 to 2025-12	645	0	0 ticker-month-model	44,828 rows per six models

The final target has mean 0.0129067, standard deviation 0.109243, median 0.0108016, 1st percentile -0.262064, 99th percentile 0.325020, minimum -0.935113, and maximum 2.920526. The target is intentionally not winsorized.

10 Modeling Framework

Linear models use StandardScaler within sklearn pipelines. Trees use raw transformed predictors. Tuning minimizes annual-fold average monthly MSE. Models are compared against a historical mean, and final models are refit on all 1990-2019 development observations before 2020-2025 prediction.

11 Model-by-Model Documentation

The historical mean is the development target average. OLS-3 uses size, book-to-market, and 12-month momentum. PLS uses one selected component. Elastic Net uses alpha 0.01 and l1 ratio 0.9 and yields zero coefficients. The selected Decision Tree has depth 3 and minimum leaf 1,000; the selected Random Forest has 200 depth-2 trees, minimum leaf 100, and square-root feature sampling. Gradient Boosting exists only in legacy archives.

12 Validation and Test Design

The code uses train 1990-2014, preliminary validation 2015-2019, development 1990-2019, and test 2020-2025. Hyperparameter grids run only on train with validation years 2005-2014. Optimized models are then checked on 2015-2019. This is leakage-resistant but differs from the active README claim that tuning folds extend through 2019.

12.1 Split counts

Sample	Rows	Months	Tickers	Dates
Train	132,336	300	578	1990-01 to 2014-12
Validation	35,630	60	618	2015-01 to 2019-12
Development	167,966	360	620	1990-01 to 2019-12
Test	44,828	72	645	2020-01 to 2025-12

13 Results

PLS is numerically best by final monthly MSE and OOS R-squared. The improvement over the historical mean is very small. Random Forest has the lowest MAE among the six. Decision Tree is decisively worst. Validation optimization helps PLS, Elastic Net, and Decision Tree relative to their fixed forms, but makes Random Forest slightly worse.

13.1 Frozen final-test comparison

Rank	Model	Monthly MSE	RMSE	MAE	OOS R2	Correlation
1	PLS	0.0137866	0.1171849	0.0801883	0.0017914	0.0438813
2	Random Forest	0.0138037	0.1172563	0.0800818	0.0005754	0.0214511
3	OLS-3	0.0138049	0.1172615	0.0801398	0.0004857	0.0256182
4	Elastic Net	0.0138120	0.1172900	0.0800774	approximately 0	undefined
5	Historical Mean	0.0138120	0.1172900	0.0800774	approximately 0	undefined
6	Decision Tree	0.0156624	0.1248378	0.0848455	-0.1328434	-0.1453403

Source: `output/models/test/final_test_model_comparison.csv`, reproduced in `output/models/interpretation/final_prediction_results.csv`. Metrics recomputed from `output/models/test/final_test_predictions.parquet` differ by no more than $7.5e-11$.

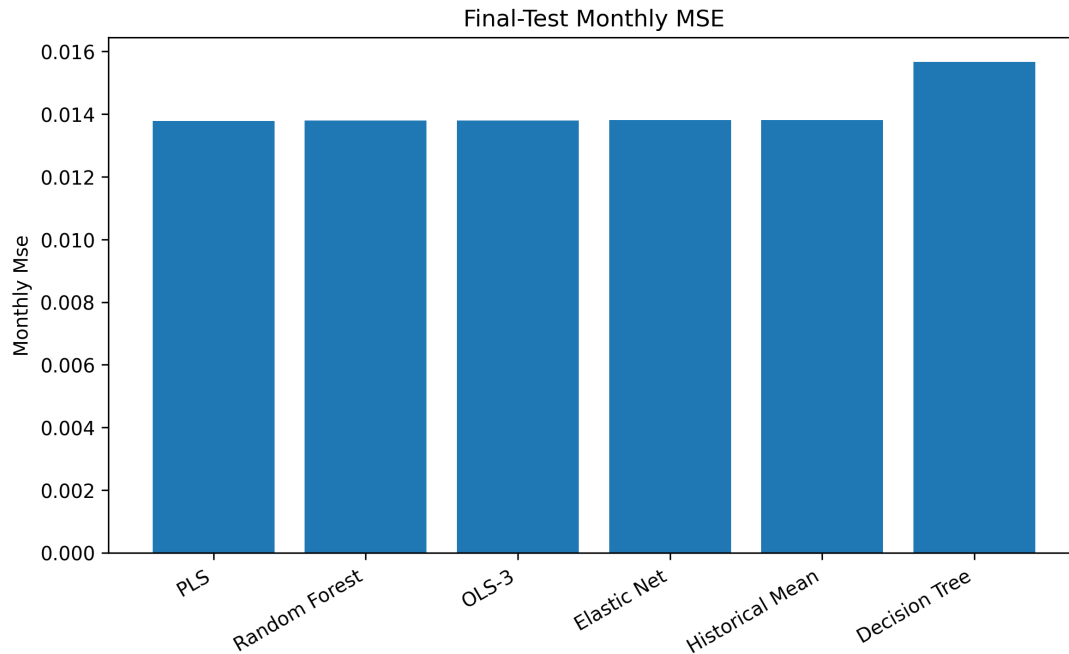


Figure 2: Final-test monthly MSE by model. Existing frozen figure; not regenerated for this audit.

14 Interpretation

Small positive OOS R-squared means a tiny reduction in squared forecast error relative to the constant benchmark, not that return variation is well explained. Negative OOS R-squared means the benchmark wins. Elastic Net's zero solution indicates the chosen penalty finds no stable incremental signal. Tree impurity importance is predictive and model-specific, not causal. PLS coefficients describe a supervised latent projection after standardization and should not be interpreted as structural premia.

15 What Was Done Well

Verified strengths include chronological sorting, month-end normalization, time-separated samples, training-only pipeline scaling, explicit historical-mean comparison, six-month accounting delay, backward as-of merging with zero violations, permanent local small inputs, saved predictor inventories, integrity checks, multiple model families, frozen interpretation artifacts, and honest negative findings.

16 Limitations and Questionable Choices

Material limitations include the 26 target continuity failures, unlagged macro merge, public ticker-based universe and matching, Yahoo revision/delisting limitations, 499-dimensional design, unwinsorized squared-loss sensitivity, weak and unstable gains, a poor single tree, zero Elastic Net solution, unpinned dependencies, stale execution wrapper, duplicate/legacy artifacts, no formal loss test, and no fully specified transaction-cost implementation in the main paper.

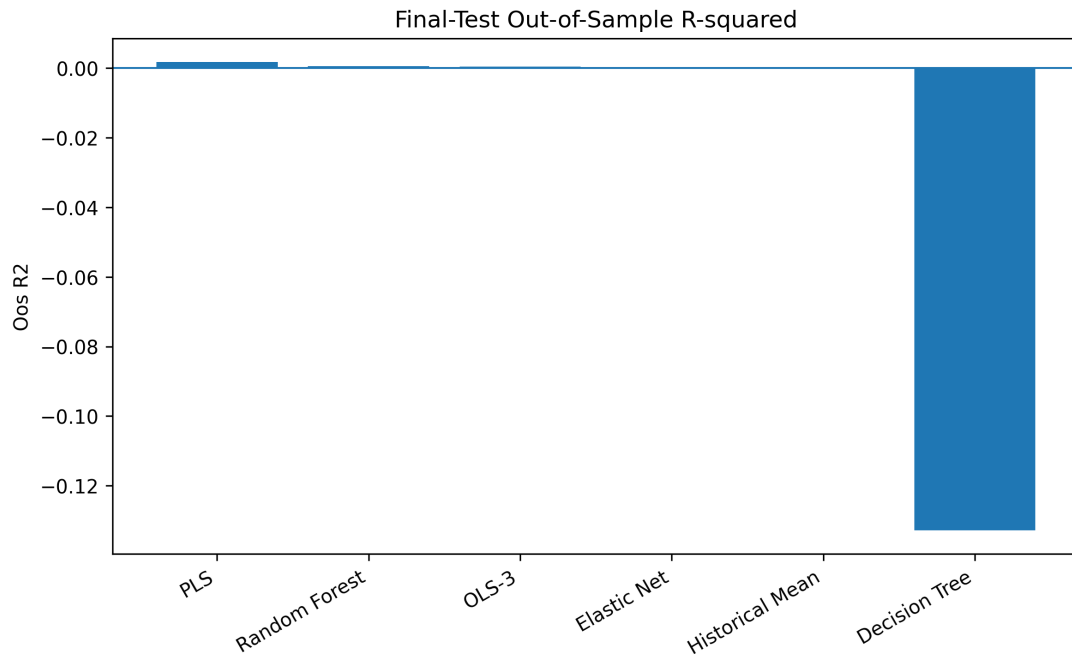


Figure 3: Final-test OOS R-squared by model. Existing frozen figure; not regenerated for this audit.

17 Leakage and Look-Ahead-Bias Audit

No direct test leakage was found in scaler fitting, hyperparameter selection, Compustat merging, or final model training. The principal timing concern is same-month macro availability. The target is future by construction, but gap transitions sometimes extend beyond one month. Human adherence to an untouched-test protocol cannot be reconstructed from files after the test outputs exist.

17.1 Verification matrix

No.	Design point	Status	Evidence
1	Next-month stock excess-return target	Partially verified	Formula is stock next-observation return minus next RF; 26 nonconsecutive transitions.
2	Time-aware shifting	Partially verified	Sorted group shift(-1), but no calendar-continuity guard.
3	Sort before group shifts	Verified	Data file 02 sorts ticker/month before shifts.
4	Predictor timing	Partially verified	Stock and Compustat timing mostly sound; macro lag is unresolved/inconsistent.
5	Month-end dates	Verified	All audited monthly inputs and panels are month-end.

No.	Design point	Status	Evidence
6	Decimal returns	Verified	RF ranges 0 to 0.0069 and return magnitudes are decimal simple returns.
7	Daily returns for volatility	Verified	Daily adjusted-close percentage changes feed monthly and rolling volatility.
8	Monthly adjusted-close return	Verified	Last monthly adjusted close percentage change.
9	Rolling features historical only	Partially verified	Momentum is lagged; contemporaneous month-end variables are used for t+1, as intended.
10	36-month history	Partially verified	min_periods=36, but the window counts observations and missing months are not explicitly checked.
11	Six-month Compustat lag	Verified	207,739 matched rows and zero saved timing violations.
12	Backward Compustat as-of merge	Verified	direction='backward' by ticker.
13	No future Compustat reports	Verified	Zero comp_available_month > stock month violations.
14	Macro variables appropriately lagged	Inconsistent	Active code merges same-month values; intended one-month lag absent.
15	Fama-French alignment	Verified	Month-end RF shifted one row; monthly input is unique and decimal.
16	Market/VIX are market-wide	Verified	At most one value per month across tickers for all four variables.
17	Predictors lagged where required	Partially verified	Target and momentum timing explicit; macro publication timing unresolved.
18	Training-only standardization	Verified	StandardScaler is inside each fitted sklearn pipeline.
19	No validation/test scaler fit	Verified	Pipeline fit occurs on fold training, train, or development arrays only.

No.	Design point	Status	Evidence
20	Chronological rather than random CV	Verified	Annual expanding masks use years < validation year.
21	Expanding folds use earlier data	Verified	1990-2004 -> 2005 through 1990-2013 -> 2014 in actual tuning.
22	Hyperparameters exclude test	Verified	Tuning loads train only; final script only reads saved parameters.
23	Test not used for tuning	Verified in code	Human viewing history is not verifiable from current repository.
24	Final test untouched until evaluation	Not verifiable	Code separates it, but procedural history cannot be proven.
25	Predictor file matches model columns	Verified	499 unique names exactly equal parquet columns after identifiers/target.
26	Identifiers and target excluded	Verified	Predictor list begins after ticker, month, target.
27	Missing handling consistent	Verified for final panel	Monthly median then zero; final missing count is zero.
28	No duplicate ticker-month rows	Verified	Zero in monthly, raw, and ranked panels.
29	Documented dataset dimensions	Verified for generated summaries	212,794 x 502 and 499 predictors match output metadata.
30	Result tables match predictions	Verified	Recomputed discrepancies below 7.5e-11.
31	Figures match frozen results	Verified by lineage/timestamps	All figures are newer than source result tables.
32	Paper values consistent	Verified for direct-linked tables	Current paper reads frozen CSV/PNG outputs and is newer than figures.
33	Random seeds consistent	Partially verified	42 is used for stochastic active models; PLS/OLS deterministic.
34	Fixed/optimized naming	Verified in outputs	Suffixes are consistent; final test removes suffixes deliberately.
35	Historical mean uses allowable data	Verified	Final constant equals development mean up to float32 precision.

18 Reproducibility Audit

Data steps are mostly explicit and canonical paths are centralized. Reproducibility is weakened by online per-ticker Yahoo downloads, an unscripted Compustat extraction, absolute manifest paths, unpinned packages, no lock file, missing shell targets, absent tests, and no recorded runtime/provenance hashes. Existing outputs permit result reproduction without retraining, but not proof that they came from the current exact source state.

19 Consistency and Staleness Audit

Final metrics recomputed from predictions agree within floating precision, the ranking order is correct, figures are newer than source result tables, and the paper is newer than figures. Inconsistencies are concentrated in workflow documentation, shell paths, duplicated input/legacy files, orphan bytecode, and intended versus actual macro/tuning logic.

19.1 Evidence-supported inconsistency register

ID	Severity	Dimension	Issue	Evidence	Recommended check
I01	High	Validity	26 targets cross gaps and are not next-calendar-month returns	Monthly panel plus data file 02 shift(-1)	Require the next observed month to equal month+1 before assigning target
I02	High	Validity	Same-month macro merge contradicts intended one-month lag	data file 04:62-84	Verify publication timing and lag macro inputs explicitly
I03	High	Reproducibility	run_pipeline.sh calls five missing scripts	Direct existence check	Update wrapper after deciding canonical stages
I04	High	Reproducibility	README tuning period and one-standard-error claim do not match code	models README versus tuning/estimation code	Correct documentation or implementation
I05	Medium	Reproducibility	Failed Yahoo downloads are printed but not persisted	data file 01 main	Save an acquisition failure report
I06	Medium	Validity	Wikipedia union cannot guarantee point-in-time membership or delisting returns	Universe acquisition logic	Use CRSP permanent identifiers for a future version
I07	Medium	Validity	Ticker-based Compustat-Yahoo merge risks symbol-change mismatches	data file 03	Audit permanent identifier mapping
I08	Medium	Interpretation	499 predictors include 376 interactions	Cleaning summary	Report regularization burden and run lower-dimensional sensitivity
I09	Medium	Interpretation	Unwinsorized target contains 91 observations above 100% absolute return	extreme_target_counts.csv	Report robustness to robust loss or documented winsorization
I10	Medium	Empirical credibility	No formal forecast-loss comparison test	No active test file found	Add Diebold-Mariano or panel-aware resampling in future
I11	Medium	Economic interpretation	Portfolio outputs exist but current paper focuses mostly on prediction	output/models/portfolio versus thesis	Decide whether portfolio evidence is in scope and discuss costs
I12	Medium	Reproducibility	requirements are unpinned and Python version is unspecified	requirements.txt	Lock package and Python versions
I13	Medium	Reproducibility	Input manifest stores machine-specific absolute paths	input/input_manifest.csv	Store repository-relative paths
I14	Medium	Staleness	Source caches remain for deleted scripts	orphan pyc inventory	Remove caches from submissions; do not treat them as source
I15	Low	Organization	Identical Fama-French RF files exist in input/ and input/external/	Matching SHA-1	Keep one canonical copy after confirming consumers
I16	Low	Organization	Large legacy and backup trees duplicate old model logic and outputs	backup/ and legacy/	Exclude from submission archive or label clearly

ID	Severity	Dimension	Issue	Evidence	Recommended check
I17	Medium	Reproducibility	No automated unit-test suite is present	Repository tree	Add tests for target continuity, lags, folds, and metrics
I18	Low	Presentation	Existing thesis build auxiliaries and OS metadata are present	thesis/*.aux etc. and .DS_Store	Clean submission artifact directories
I19	Medium	Data quality	54 infinities exist before final cleaning	Dataset scan	Document affected columns and assert replacement before imputation
I20	Medium	Methodology	Beta/idio-vol formula is a rolling variance identity, not an explicit regression with intercept	data file 02:306-321	Clarify definition or compare with rolling OLS
I21	Low	Presentation	Existing bibliography contains an unresolved course-reference comment	thesis/references.bib	Verify required course citations before submission
I22	Medium	Process	Untouched-test discipline cannot be proven from code after results exist	Repository state	Describe the procedural claim cautiously

20 Required Checks Before Submission

The priority is to verify target calendar continuity and macro publication timing, repair the run instructions, align the methodological description with actual folds and selection rule, confirm required citations, state the universe/ticker limitations, and decide whether the portfolio evidence is in scope. These checks should be completed before claiming a fully reproducible and leakage-safe submission.

21 Recommended Future Improvements

Use point-in-time CRSP/Compustat identifiers and delisting returns, actual filing dates, explicit macro release lags, a calendar-grid target, locked environments, automated tests, provenance hashes, nested or repeated chronological validation, robust-loss sensitivity, formal forecast-comparison inference, and a prespecified net-of-cost portfolio design.

22 Final Overall Assessment

The project is methodologically thoughtful and unusually transparent for a course project, but it requires important checks before submission. The core result - weak predictability with several model failures - is credible as a negative finding. Strong claims about a strict one-month target, macro timing, complete reproducibility, or profitability are not yet justified.

22.1 Scorecard

Category	Score / 10	Reason
Methodological quality	7	Strong chronology and benchmarks; target continuity and macro lag require correction
Data quality	6	Large transparent panel; public-vendor and ticker-history limitations
Leakage protection	8	Scaler, test, and Compustat protections verified; macro timing unresolved
Reproducibility	5	Frozen outputs and central config help; wrapper, versions, and Compustat extraction hinder
Code organization	7	Clear stages/utilities; archives, stale paths, and caches create ambiguity
Documentation	6	Substantial guides and paper; tuning claims drift from code
Empirical credibility	7	Negative findings are honest; no formal loss test and gains are tiny
Submission readiness	6	Requires important checks before submission

Final verdict: Requires important checks.

22.1.1 Top ten strongest aspects

1. Chronological train/validation/development/test separation.
2. Training-only pipeline standardization.
3. Verified six-month Compustat lag.
4. Verified backward as-of merge.
5. Permanent local small external inputs.
6. Exact predictor inventory.
7. Multiple transparent benchmarks and model families.
8. Frozen predictions and recomputable metrics.
9. Honest reporting of weak and negative results.
10. Interpretation, diagnostics, robustness, and portfolio artifacts.

22.1.2 Top ten weaknesses

1. Twenty-six non-calendar target transitions.
2. Same-month macro merge.
3. Stale shell workflow.
4. Tuning documentation mismatch.
5. Unpinned environment.
6. Public ticker universe and matching.
7. Unscripted Compustat extraction.
8. High interaction dimensionality.
9. No formal forecast-loss test.
10. No proof of historical untouched-test discipline.

23 Reproduction Guide

Three reproducibility levels are possible: rebuild all data (requires network access and the supplied Compustat CSV), rerun models from the final parquet, or regenerate interpretation and paper artifacts only. Because `run_pipeline.sh` is stale, the canonical manual order in this report should be followed until the wrapper is corrected.

23.1 Required environment

The repository lists `pandas`, `numpy`, `scikit-learn`, `statsmodels`, `requests`, `yfinance`, `pyarrow`, `matplotlib`, `openpyxl`. The inspected environment is Python 3.13.2 with `pandas` 2.3.3, `numpy` 2.2.3, `scikit-learn` 1.8.0, `statsmodels` 0.14.5, `requests` 2.32.3, `yfinance` 1.4.1, `pyarrow` 19.0.1, `matplotlib` 3.10.1, and `openpyxl` 3.1.5. These are observations, not locked requirements. WRDS credentials are required only to recreate the Compustat extraction, whose query is absent. Wikipedia/Yahoo/Ken French access is required for acquisition. No FRED API is used.

23.2 Three run levels

Raw rebuild: run acquisition scripts as needed, provide raw Compustat, then data files 01-05. This downloads large stock histories and was not executed during this audit.

Model-only rebuild: with the final parquet and predictor CSV present, run fixed, tuning, optimized comparison, robustness, final test, then interpretation using the active paths documented above. Do not rely on the current shell wrapper.

Reporting-only rebuild: run `python src/models/step_06_interpretation/run_all.py` only if frozen test/validation inputs are already present, then compile `thesis/paper.tex` with its Makefile.

24 Glossary

Key terms: adjusted close (split/dividend-adjusted price); OOS R2 (benchmark-relative squared-error improvement); monthly MSE (equal-weighted mean of within-month MSE); PLS (partial least squares); RF (risk-free rate or, when named as a model, Random Forest); SIC2 (two-digit industry code); expanding window (all prior years train the next year); development sample (all pre-test observations).

A Full Directory Tree

The tree is a complete pre-audit snapshot excluding `.git` and the newly created audit directory.

```
.DS_Store
.gitignore
.matplotlib/fontlist-v390.json
README.md
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docs/Data_Construction_Guide.tex
docs/Pipeline_Overview.pdf
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```

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src/data/__pycache__/04_build_raw_kelly_dataset.cpython-313.pyc
src/data/__pycache__/05_clean_and_rank_normalize.cpython-313.pyc
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```
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src/models/step_01_fixed/__pycache__/04_diagnose_fixed_models.cpython-313.pyc
src/models/step_01_fixed/__pycache__/__init__.cpython-313.pyc
src/models/step_02_tuning/01_tune_linear_models.py
src/models/step_02_tuning/02_tune_tree_models.py
src/models/step_02_tuning/03_compare_tuning_results.py
src/models/step_02_tuning/__init__.py
src/models/step_02_tuning/__pycache__/01_tune_linear_models.cpython-313.pyc
src/models/step_02_tuning/__pycache__/02_tune_tree_models.cpython-313.pyc
src/models/step_02_tuning/__pycache__/03_compare_scaling.cpython-313.pyc
src/models/step_02_tuning/__pycache__/03_compare_tuning_results.cpython-313.pyc
src/models/step_02_tuning/__pycache__/__init__.cpython-313.pyc
src/models/step_03_optimization/01_optimized_linear_models.py
src/models/step_03_optimization/02_optimized_tree_models.py
src/models/step_03_optimization/03_compare_optimized_models.py
src/models/step_03_optimization/04_compare_fixed_vs_optimized.py
src/models/step_03_optimization/__init__.py
src/models/step_03_optimization/__pycache__/01_optimized_linear_models.cpython-313.pyc
src/models/step_03_optimization/__pycache__/02_optimized_tree_models.cpython-313.pyc
src/models/step_03_optimization/__pycache__/03_compare_optimized_models.cpython-313.pyc
src/models/step_03_optimization/__pycache__/04_compare_fixed_vs_optimized.cpython-313.pyc
src/models/step_03_optimization/__pycache__/05_time_series_robustness.cpython-313.pyc
src/models/step_03_optimization/__pycache__/06_final_test_evaluation.cpython-313.pyc
src/models/step_03_optimization/__pycache__/__init__.cpython-313.pyc
src/models/step_04_robustness/01_time_series_robustness.py
src/models/step_04_robustness/__init__.py
src/models/step_04_robustness/__pycache__/01_time_series_robustness.cpython-313.pyc
src/models/step_04_robustness/__pycache__/__init__.cpython-313.pyc
src/models/step_05_test/01_final_test_evaluation.py
src/models/step_05_test/__init__.py
src/models/step_05_test/__pycache__/01_final_test_evaluation.cpython-313.pyc
src/models/step_05_test/__pycache__/__init__.cpython-313.pyc
src/models/step_06_interpretation/00_utils.py
src/models/step_06_interpretation/01_model_interpretation.py
src/models/step_06_interpretation/02_feature_importance.py
src/models/step_06_interpretation/03_create_result_tables.py
src/models/step_06_interpretation/04_create_figures.py
src/models/step_06_interpretation/05_report_notes.py
src/models/step_06_interpretation/README.md
src/models/step_06_interpretation/__init__.py
src/models/step_06_interpretation/__pycache__/00_utils.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/01_model_interpretation.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/02_feature_importance.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/03_create_result_tables.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/04_create_figures.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/05_report_notes.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/__init__.cpython-313.pyc
src/models/step_06_interpretation/__pycache__/run_all.cpython-313.pyc
src/models/step_06_interpretation/run_all.py
src/models/utils/__init__.py
src/models/utils/__pycache__/__init__.cpython-313.pyc
src/models/utils/__pycache__/data.cpython-313.pyc
src/models/utils/__pycache__/estimation.cpython-313.pyc
src/models/utils/__pycache__/evaluation.cpython-313.pyc
src/models/utils/__pycache__/portfolio.cpython-313.pyc
src/models/utils/data.py
src/models/utils/estimation.py
src/models/utils/evaluation.py
thesis/Makefile
thesis/README.md
thesis/paper.aux
thesis/paper.bbl
thesis/paper.bcf
thesis/paper.fdb_latexmk
thesis/paper.fls
thesis/paper.log
thesis/paper.out
thesis/paper.pdf
thesis/paper.run.xml
thesis/paper.tex
thesis/references.bib
```

B Complete File Inventory

B.1 Identity and classification inventory

Path	Type	Size	Class	Role	Status
.DS_Store	.DS_Store	10.0 KB	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
.gitignore	.gitignore	632.0 B	configuration or environment	Repository support file; exact role follows from its category and name.	Review
.matplotlib/fontlist-v390.json	.json	220.7 KB	other	Repository support file; exact role follows from its category and name.	Review
README.md	.md	2.4 KB	documentation	Repository support file; exact role follows from its category and name.	Review
backup/model_reorganization_20260719_054538/audit_report.txt	.txt	6.0 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/existing_output_files.txt	.txt	3.9 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/original_directory_tree.txt	.txt	902.0 B	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/01_fixed_linear_models.py	.py	3.3 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/02_fixed_tree_models.py	.py	4.2 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/03_compare_fixed_models.py	.py	1.2 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/01_optimized_linear_models.py	.py	3.9 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/02_optimized_tree_models.py	.py	2.7 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/03_compare_optimized_models.py	.py	1.5 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	.py	5.0 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	.py	3.6 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	.py	2.9 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/gradient_boosting_inventory.txt	.txt	3.1 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/README.md	.md	1.7 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/README_MODEL_WORKFLOW.md	.md	2.8 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/02_fixed_tree_models.py	.py	2.9 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/03_compare_fixed_models.py	.py	1.4 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/02_tune_tree_models.py	.py	3.1 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/03_compare_tuning_results.py	.py	1.3 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/02_optimized_tree_models.py	.py	3.8 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/03_compare_optimized_models.py	.py	1.7 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	.py	5.6 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_04_robustness/01_time_series_robustness.py	.py	4.2 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_05_test/01_final_test_evaluation.py	.py	7.0 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/01_portfolio_sorts.py	.py	3.1 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/02_compare_portfolio_results.py	.py	1.8 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/03_final_model_and_portfolio_summary.py	.py	1.5 KB	backup or possibly obsolete	Archived snapshot; not part of the active workflow.	Possibly obsolete
docs/Data_Construction_Guide.pdf	.pdf	252.7 KB	documentation	Earlier data-construction or pipeline documentation.	Review
docs/Data_Construction_Guide.tex	.tex	23.8 KB	documentation	Earlier data-construction or pipeline documentation.	Review
docs/Pipeline_Overview.pdf	.pdf	168.1 KB	documentation	Earlier data-construction or pipeline documentation.	Review
docs/figures/.gitkeep	.gitkeep	1.0 B	documentation	Earlier data-construction or pipeline documentation.	Review
input/.DS_Store	.DS_Store	8.0 KB	cache or temporary	Permanent local input or input metadata.	Cache/temporary
input/README.md	.md	594.0 B	data input	Permanent local input or input metadata.	Input
input/external/.DS_Store	.DS_Store	6.0 KB	cache or temporary	Permanent local input or input metadata.	Cache/temporary
input/external/.gitkeep	.gitkeep	0.0 B	data input	Permanent local input or input metadata.	Input
input/external/fama_french_rf_monthly.csv	.csv	21.5 KB	data input	Permanent local input or input metadata.	Input
input/external/welch_goyal_macro_1990_2025.csv	.csv	58.8 KB	data input	Permanent local input or input metadata.	Input

Path	Type	Size	Class	Role	Status
input/fama_french_rf_monthly.csv	.csv	21.5 KB	data input	Permanent local input or input metadata.	Input
input/input_manifest.csv	.csv	1.2 KB	data input	Permanent local input or input metadata.	Input
input/market_gspc_daily.csv	.csv	1.0 MB	data input	Permanent local input or input metadata.	Input
input/market_vix_daily.csv	.csv	916.6 KB	data input	Permanent local input or input metadata.	Input
input/raw/.DS_Store	.DS_Store	6.0 KB	cache or temporary	Permanent local input or input metadata.	Cache/temporary
input/raw/.gitkeep	.gitkeep	0.0 B	data input	Permanent local input or input metadata.	Input
input/raw/compustat_annual_1980_2025.csv	.csv	13.8 MB	data input	Permanent local input or input metadata.	Input
input/stock_universe_locke_d.csv	.csv	3.5 KB	data input	Permanent local input or input metadata.	Input
output/data/final/cleaning_summary.csv	.csv	725.0 B	final data	Saved final or model-ready data product.	Frozen output
output/data/final/daily_prices_clean_1987_2026.csv	.csv	475.5 MB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/dropped_predictors_empty.csv	.csv	17.0 B	final data	Saved final or model-ready data product.	Frozen output
output/data/final/extreme_target_counts.csv	.csv	151.0 B	final data	Saved final or model-ready data product.	Frozen output
output/data/final/extreme_target_observations.csv	.csv	4.7 KB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/kelly_raw_dataset_summary.csv	.csv	312.0 B	final data	Saved final or model-ready data product.	Frozen output
output/data/final/model_dataset_kelly_ranked_full_1990_2025.parquet	.parquet	414.0 MB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/model_dataset_kelly_raw_full_1990_2025.csv	.csv	229.1 MB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/monthly_imputation_medians_summary.csv	.csv	7.2 KB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/monthly_stock_panel_with_targets_1990_2025.csv	.csv	109.2 MB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/predictor_columns_kelly_ranked.csv	.csv	8.9 KB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/predictor_columns_kelly_raw.csv	.csv	1.2 KB	final data	Saved final or model-ready data product.	Frozen output
output/data/final/removed_tickers.csv	.csv	602.0 B	final data	Saved final or model-ready data product.	Frozen output
output/data/final/sp500_tickers_clean.csv	.csv	2.7 KB	final data	Saved final or model-ready data product.	Frozen output

Path	Type	Size	Class	Role	Status
output/data/intermediate/compustat_annual_cleaned_1980_2025.csv	.csv	9.7 MB	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/daily_prices_raw_1987_2026.csv	.csv	476.0 MB	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/monthly_panel_with_compustat_macro_1990_2025.csv	.csv	200.3 MB	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/monthly_removed_tickers.csv	.csv	684.0 B	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/monthly_return_quality_report.csv	.csv	50.7 KB	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/sp500_tickers_raw.csv	.csv	3.5 KB	intermediate data	Saved intermediate data product.	Frozen output
output/data/intermediate/ticker_quality_report.csv	.csv	61.7 KB	intermediate data	Saved intermediate data product.	Frozen output
output/models/fixed/diagnostics/decision_tree_feature_importance.csv	.csv	11.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/decision_tree_rules.txt	.txt	271.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/decision_tree_split_feature_distributions.csv	.csv	1.2 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/decision_tree_structure.csv	.csv	611.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/decision_tree_train_leaf_distribution.csv	.csv	222.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/decision_tree_validation_leaf_distribution.csv	.csv	83.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/diagnostic_warning_summary.csv	.csv	2.1 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/diagnostic_warnings.txt	.txt	227.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/extreme_predictions.csv	.csv	33.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/fixed_data_summary.csv	.csv	489.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/fixed/diagnostics/fixed_model_performance_by_month.csv	.csv	63.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/fixed_model_performance_by_validation_year.csv	.csv	8.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/fixed_prediction_distribution.csv	.csv	3.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/largest_absolute_prediction_rows.csv	.csv	300.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/largest_absolute_target_rows.csv	.csv	70.2 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/largest_prediction_rows.csv	.csv	287.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/largest_target_rows.csv	.csv	67.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/ols3_predictor_distribution.csv	.csv	1005.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/ols3_selected_predictors.csv	.csv	46.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/prediction_summary.csv	.csv	2.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/predictor_distribution_flags.csv	.csv	23.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/predictor_distribution_summary.csv	.csv	152.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/random_forest_extreme_predictions.csv	.csv	22.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/random_forest_feature_importance.csv	.csv	15.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/random_forest_prediction_summary.csv	.csv	1.1 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/smallest_prediction_rows.csv	.csv	321.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/fixed/diagnostics/smallest_target_rows.csv	.csv	74.9 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/squared_error_concentration.csv	.csv	6.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/target_extreme_counts.csv	.csv	783.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/target_quantiles.csv	.csv	1.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/target_summary.csv	.csv	400.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/target_summary_by_month.csv	.csv	41.2 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/target_summary_by_year.csv	.csv	4.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/diagnostics/validation_performance_by_year.csv	.csv	7.1 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_all_model_metrics.csv	.csv	2.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_linear_model_coefficients.csv	.csv	49.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_linear_model_metrics.csv	.csv	1.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_linear_model_predictions.parquet	.parquet	9.0 MB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_model_summary_wide.csv	.csv	1.2 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_model_validation_ranking.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_tree_model_feature_importance.csv	.csv	68.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/fixed/fixed_tree_model_metrics.csv	.csv	1.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/fixed/fixed_tree_model_predictions.parquet	.parquet	5.2 MB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/best_hyperparameters.csv	.csv	422.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/decision_tree_feature_importance.csv	.csv	19.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/elastic_net_coefficients.csv	.csv	23.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/feature_importance_top_variables.csv	.csv	4.9 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/prediction_distribution_by_model.png	.png	89.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/prediction_vs_realized_pls.png	.png	112.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/test_correlation_by_model.png	.png	97.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/test_monthly_mse_by_model.png	.png	114.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/test_oos_r2_by_model.png	.png	108.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/top_decision_tree_features.png	.png	136.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/top_elastic_net_coefficients_skipped.txt	.txt	189.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/top_pls_coefficients.png	.png	215.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/top_random_forest_features.png	.png	185.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/yearly_mse_by_model.png	.png	170.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/figures/yearly_oos_r2_by_model.png	.png	148.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/interpretation/final_model_complexity.csv	.csv	438.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/final_prediction_results.csv	.csv	669.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/final_report_results.xlsx	.xlsx	17.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/fixed_vs_optimized_results.csv	.csv	1.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/model_behavior_flags.csv	.csv	595.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/model_complexity_summary.csv	.csv	409.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/model_interpretation_metrics.csv	.csv	2.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/ols_3_coefficients.csv	.csv	254.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/pls_coefficients.csv	.csv	36.9 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/pls_components.csv	.csv	113.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/random_forest_feature_importance.csv	.csv	21.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/report_notes.md	.md	7.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/tree_model_complexity.csv	.csv	353.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/interpretation/yearly_prediction_results.csv	.csv	4.1 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/legacy/gradient_boosting/mixed/fixed_diagnostics/decision_tree_feature_importance.csv	.csv	11.7 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_split_feature_distributions.csv	.csv	1.2 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_structure.csv	.csv	611.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_train_leaf_distribution.csv	.csv	222.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_validation_leaf_distribution.csv	.csv	83.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/diagnostic_warning_summary.csv	.csv	2.1 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/extreme_predictions.csv	.csv	33.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_data_summary.csv	.csv	489.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_model_performance_by_month.csv	.csv	63.5 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_model_performance_by_validation_year.csv	.csv	8.3 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_prediction_distribution.csv	.csv	3.4 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/gradient_boosting_extreme_predictions.csv	.csv	24.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/gradient_boosting_feature_importance.csv	.csv	12.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_absolute_prediction_rows.csv	.csv	300.7 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_absolute_target_rows.csv	.csv	70.2 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_prediction_rows.csv	.csv	287.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_target_rows.csv	.csv	67.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/ols3_predictor_distribution.csv	.csv	1005.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/ols3_selected_predictors.csv	.csv	46.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/prediction_summary.csv	.csv	2.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/predictor_distribution_flags.csv	.csv	23.4 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/predictor_distribution_summary.csv	.csv	152.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_extreme_predictions.csv	.csv	22.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_feature_importance.csv	.csv	15.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_prediction_summary.csv	.csv	1.1 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/smallest_prediction_rows.csv	.csv	321.4 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/smallest_target_rows.csv	.csv	74.9 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/squared_error_concentration.csv	.csv	6.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_extreme_counts.csv	.csv	783.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_quantiles.csv	.csv	1.7 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary.csv	.csv	400.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary_by_month.csv	.csv	41.2 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary_by_year.csv	.csv	4.5 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/validation_performance_by_year.csv	.csv	7.1 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/all_model_metrics.csv	.csv	3.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed/model_summary_wide.csv	.csv	1.2 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
output/models/legacy/gradient_boosting/mixed/fixed_model_validation_ranking.csv	.csv	1.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed_tree_model_feature_importance.csv	.csv	68.7 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed_tree_model_metrics.csv	.csv	1.5 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/fixed_tree_model_predictions.parquet	.parquet	5.2 MB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_all_metrics.csv	.csv	2.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_family_summary.csv	.csv	1.2 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_validation_ranking.csv	.csv	2.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/optimized_all_model_metrics.csv	.csv	3.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/optimized_model_validation_ranking.csv	.csv	1.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_feature_importance.csv	.csv	72.3 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_metrics.csv	.csv	1.5 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_predictions.parquet	.parquet	5.1 MB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
output/models/legacy/gradient_boosting/mixed/tuning/tuning_all_results.csv	.csv	11.3 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/mixed/tuning/tuning_summary.csv	.csv	652.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/standalone/gradient_boosting_best_parameters.csv	.csv	129.0 B	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/standalone/gradient_boosting_extreme_predictions.csv	.csv	24.8 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/standalone/gradient_boosting_feature_importance.csv	.csv	12.6 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/legacy/gradient_boosting/standalone/gradient_boosting_tuning_results.csv	.csv	5.0 KB	model output	Archived snapshot; not part of the active workflow.	Possibly obsolete
output/models/optimization/fixed_vs_optimized_all_metrics.csv	.csv	2.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/fixed_vs_optimized_family_summary.csv	.csv	982.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/fixed_vs_optimized_validation_ranking.csv	.csv	2.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_all_model_metrics.csv	.csv	2.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_linear_model_coefficients.csv	.csv	48.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_linear_model_metrics.csv	.csv	1.7 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_linear_model_predictions.parquet	.parquet	8.0 MB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_model_validation_ranking.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_tree_model_feature_importance.csv	.csv	72.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/optimization/optimized_tree_model_metrics.csv	.csv	1.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/optimization/optimized_tree_model_predictions.parquet	.parquet	5.1 MB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/cumulative_long_short_returns.png	.png	317.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/final_portfolio_report_table.csv	.csv	1.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/final_prediction_portfolio_ranking.csv	.csv	4.2 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/final_prediction_portfolio_summary.csv	.csv	4.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_annual_performance.csv	.csv	2.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_block_bootstrap.csv	.csv	777.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_cumulative_returns.csv	.csv	22.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_drawdown_statistics.csv	.csv	284.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_monthly_returns.csv	.csv	36.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_ranking.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/portfolio/portfolio_summary.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/robustness/time_series_robustness_metrics.csv	.csv	659.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/robustness/time_series_robustness_predictions.parquet	.parquet	746.8 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/robustness/time_series_robustness_yearly_metrics.csv	.csv	2.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/test/final_test_metrics.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/final_test_model_comparison.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/final_test_predictions.parquet	.parquet	1.8 MB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/interpretation_inputs/linear_model_coefficients.csv	.csv	60.9 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/interpretation_inputs/model_completeness.csv	.csv	411.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/interpretation_inputs/pls_components.csv	.csv	113.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/test/interpretation_inputs/tree_feature_importance.csv	.csv	41.6 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/decision_tree_best_parameters.csv	.csv	83.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/decision_tree_tuning_results.csv	.csv	1.4 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/elastic_net_best_parameters.csv	.csv	71.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/elastic_net_tuning_results.csv	.csv	1.3 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/pls_best_parameters.csv	.csv	48.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/pls_tuning_results.csv	.csv	521.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/random_forest_best_parameters.csv	.csv	127.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/random_forest_tuning_results.csv	.csv	2.5 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/models/tuning/tuning_all_results.csv	.csv	6.0 KB	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output

Path	Type	Size	Class	Role	Status
output/models/tuning/tuning_summary.csv	.csv	486.0 B	model output	Frozen model, diagnostic, portfolio, or interpretation output.	Frozen output
output/reports/kelly_characteristic_comparison.csv	.csv	8.8 KB	other	Repository support file; exact role follows from its category and name.	Frozen output
requirements.txt	.txt	84.0 B	configuration or environment	Unpinned Python dependency list.	Active or support
run_pipeline.sh	.sh	1.4 KB	configuration or environment	Intended end-to-end execution wrapper; currently stale after model reorganization.	Active or support
src/.DS_Store	.DS_Store	6.0 KB	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/__init__.py	.py	0.0 B	source code	Repository support file; exact role follows from its category and name.	Active or support
src/__pycache__/__init__.cpython-313.pyc	.pyc	181.0 B	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/__pycache__/config.python-313.pyc	.pyc	3.8 KB	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/__pycache__/feature_definitions.cpython-313.pyc	.pyc	1.0 KB	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/acquisition/01_create_locked_stock_universe.py	.py	3.5 KB	source code	One-time acquisition or input-registration logic.	Active or support
src/acquisition/02_create_fama_french_rf_input.py	.py	2.2 KB	source code	One-time acquisition or input-registration logic.	Active or support
src/acquisition/03_download_market_inputs.py	.py	2.3 KB	source code	One-time acquisition or input-registration logic.	Active or support
src/acquisition/04_create_welch_goyal_input.py	.py	2.2 KB	source code	One-time acquisition or input-registration logic.	Active or support
src/acquisition/__init__.py	.py	48.0 B	source code	One-time acquisition or input-registration logic.	Active or support
src/acquisition/__pycache__/01_create_locked_stock_universe.cpython-313.pyc	.pyc	6.0 KB	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary
src/acquisition/__pycache__/02_create_fama_french_rf_input.cpython-313.pyc	.pyc	4.2 KB	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary
src/acquisition/__pycache__/03_download_market_inputs.cpython-313.pyc	.pyc	3.9 KB	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary
src/acquisition/__pycache__/04_create_welch_goyal_input.cpython-313.pyc	.pyc	3.4 KB	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary
src/acquisition/__pycache__/__init__.cpython-313.pyc	.pyc	180.0 B	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary

Path	Type	Size	Class	Role	Status
src/acquisition/.__pycache__/_manifest.cpython-313.pyc	.pyc	1.7 KB	cache or temporary	One-time acquisition or input-registration logic.	Cache/temporary
src/acquisition/manifest.py	.py	1.2 KB	source code	One-time acquisition or input-registration logic.	Active or support
src/config.py	.py	4.2 KB	source code	Central paths, dates, target, and sample boundaries.	Active or support
src/data/.DS_Store	.DS_Store	6.0 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/01_build_clean_yahoo_daily.py	.py	7.2 KB	source code	Active data-construction stage.	Active or support
src/data/02_build_monthly_stock_features.py	.py	12.1 KB	source code	Active data-construction stage.	Active or support
src/data/03_add_fundamentals_and_macro.py	.py	11.1 KB	source code	Active data-construction stage.	Active or support
src/data/04_build_raw_kelly_dataset.py	.py	5.4 KB	source code	Active data-construction stage.	Active or support
src/data/05_clean_and_rank_normalize.py	.py	10.3 KB	source code	Active data-construction stage.	Active or support
src/data/.__init__.py	.py	0.0 B	source code	Active data-construction stage.	Active or support
src/data/.__pycache__/01_build_clean_yahoo_daily.cpython-313.pyc	.pyc	9.0 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/02_build_monthly_stock_features.cpython-313.pyc	.pyc	14.4 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/03_add_fundamentals_and_macro.cpython-313.pyc	.pyc	13.1 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/04_build_raw_kelly_dataset.cpython-313.pyc	.pyc	9.3 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/05_clean_and_rank_normalize.cpython-313.pyc	.pyc	11.7 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/06_inspect_final_dataset.cpython-313.pyc	.pyc	16.8 KB	cache or temporary	Active data-construction stage.	Cache/temporary
src/data/.__pycache__/.__init__.cpython-313.pyc	.pyc	117.0 B	cache or temporary	Active data-construction stage.	Cache/temporary
src/documentation/latex/Pipeline_Overview.aux	.aux	162.0 B	other	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.fdb_latexmk	.fdb_latexmk	21.9 KB	other	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.fls	.fls	22.6 KB	other	Earlier data-construction or pipeline documentation.	Active or support

Path	Type	Size	Class	Role	Status
src/documentation/latex/Pipeline_Overview.log	.log	31.3 KB	other	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.out	.out	0.0 B	other	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.pdf	.pdf	168.1 KB	documentation	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.synctex.gz	.gz	11.9 KB	other	Earlier data-construction or pipeline documentation.	Active or support
src/documentation/latex/Pipeline_Overview.tex	.tex	4.7 KB	documentation	Earlier data-construction or pipeline documentation.	Active or support
src/feature_definitions.py	.py	1.6 KB	source code	Locked predictor-family definitions.	Active or support
src/models/.DS_Store	.DS_Store	6.0 KB	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/models/README.md	.md	1.8 KB	documentation	Repository support file; exact role follows from its category and name.	Active or support
src/models/README_MODEL_WORKFLOW.md	.md	3.0 KB	documentation	Repository support file; exact role follows from its category and name.	Active or support
src/models/__init__.py	.py	0.0 B	source code	Repository support file; exact role follows from its category and name.	Active or support
src/models/__pycache__/__init__.cpython-313.pyc	.pyc	188.0 B	cache or temporary	Machine-generated cache or operating-system metadata.	Cache/temporary
src/models/legacy/gradient_boosting/source_snapshots/README.md	.md	1.7 KB	documentation	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradient_boosting/source_snapshots/README_MODEL_WORKFLOW.md	.md	2.8 KB	documentation	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradient_boosting/source_snapshots/step_01_fixed/02_fixed_tree_models.py	.py	2.9 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/02_tune_tree_models.py	.py	3.1 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/03_compare_tuning_results.py	.py	1.3 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradient_boosting/source_snapshots/step_03_optimization/02_optimized_tree_models.py	.py	3.8 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete

Path	Type	Size	Class	Role	Status
src/models/legacy/gradien t_boosting/source_snapshots /step_04_robustness/01_tim e_series_robustness.py	.py	4.2 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/legacy/gradien t_boosting/source_snapshots /step_05_test/01_final_tes t_evaluation.py	.py	7.0 KB	source code	Archived snapshot; not part of the active workflow.	Possibly obsolete
src/models/step_01_fixed/0 1_fixed_linear_models.py	.py	3.9 KB	source code	Fixed train/validation benchmark stage.	Active or support
src/models/step_01_fixed/0 2_fixed_tree_models.py	.py	2.7 KB	source code	Fixed train/validation benchmark stage.	Active or support
src/models/step_01_fixed/0 3_compare_fixed_models.py	.py	1.6 KB	source code	Fixed train/validation benchmark stage.	Active or support
src/models/step_01_fixed/0 4_diagnose_fixed_models.py	.py	5.0 KB	source code	Fixed train/validation benchmark stage.	Active or support
src/models/step_01_fixed/_ _init__.py	.py	0.0 B	source code	Fixed train/validation benchmark stage.	Active or support
src/models/step_01_fixed/_ _pycache__/01_fixed_linear _models.cpython-313.pyc	.pyc	4.8 KB	cache or temporary	Fixed train/validation benchmark stage.	Cache/temporary
src/models/step_01_fixed/_ _pycache__/02_fixed_tree_m odels.cpython-313.pyc	.pyc	3.8 KB	cache or temporary	Fixed train/validation benchmark stage.	Cache/temporary
src/models/step_01_fixed/_ _pycache__/03_compare_fixe d_models.cpython-313.pyc	.pyc	2.3 KB	cache or temporary	Fixed train/validation benchmark stage.	Cache/temporary
src/models/step_01_fixed/_ _pycache__/04_diagnose_fix ed_models.cpython-313.pyc	.pyc	6.7 KB	cache or temporary	Fixed train/validation benchmark stage.	Cache/temporary
src/models/step_01_fixed/_ _pycache__/__init__.cpytho n-313.pyc	.pyc	133.0 B	cache or temporary	Fixed train/validation benchmark stage.	Cache/temporary
src/models/step_02_tuning/ 01_tune_linear_models.py	.py	2.5 KB	source code	Annual expanding-window tuning stage.	Active or support
src/models/step_02_tuning/ 02_tune_tree_models.py	.py	2.6 KB	source code	Annual expanding-window tuning stage.	Active or support
src/models/step_02_tuning/ 03_compare_tuning_results. py	.py	1.3 KB	source code	Annual expanding-window tuning stage.	Active or support
src/models/step_02_tuning/ __init__.py	.py	0.0 B	source code	Annual expanding-window tuning stage.	Active or support
src/models/step_02_tuning/ __pycache__/01_tune_linear _models.cpython-313.pyc	.pyc	2.9 KB	cache or temporary	Annual expanding-window tuning stage.	Cache/temporary
src/models/step_02_tuning/ __pycache__/02_tune_tree_m odels.cpython-313.pyc	.pyc	2.6 KB	cache or temporary	Annual expanding-window tuning stage.	Cache/temporary

Path	Type	Size	Class	Role	Status
src/models/step_02_tuning/____pycache____/03_compare_scaling.cpython-313.pyc	.pyc	3.8 KB	cache or temporary	Annual expanding-window tuning stage.	Cache/temporary
src/models/step_02_tuning/____pycache____/03_compare_tuning_results.cpython-313.pyc	.pyc	2.1 KB	cache or temporary	Annual expanding-window tuning stage.	Cache/temporary
src/models/step_02_tuning/____pycache____/____init____.cpython-313.pyc	.pyc	134.0 B	cache or temporary	Annual expanding-window tuning stage.	Cache/temporary
src/models/step_03_optimization/01_optimized_linear_models.py	.py	4.5 KB	source code	Optimized train/validation comparison stage.	Active or support
src/models/step_03_optimization/02_optimized_tree_models.py	.py	3.3 KB	source code	Optimized train/validation comparison stage.	Active or support
src/models/step_03_optimization/03_compare_optimized_models.py	.py	1.9 KB	source code	Optimized train/validation comparison stage.	Active or support
src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	.py	6.0 KB	source code	Optimized train/validation comparison stage.	Active or support
src/models/step_03_optimization/____init____.py	.py	0.0 B	source code	Optimized train/validation comparison stage.	Active or support
src/models/step_03_optimization/____pycache____/01_optimized_linear_models.cpython-313.pyc	.pyc	5.3 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_03_optimization/____pycache____/02_optimized_tree_models.cpython-313.pyc	.pyc	4.4 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_03_optimization/____pycache____/03_compare_optimized_models.cpython-313.pyc	.pyc	2.5 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_03_optimization/____pycache____/04_compare_fixed_vs_optimized.cpython-313.pyc	.pyc	5.2 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_03_optimization/____pycache____/05_time_series_robustness.cpython-313.pyc	.pyc	5.9 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_03_optimization/____pycache____/06_final_test_evaluation.cpython-313.pyc	.pyc	7.2 KB	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary

Path	Type	Size	Class	Role	Status
src/models/step_03_optimization/__pycache__/__init__.cpython-313.pyc	.pyc	140.0 B	cache or temporary	Optimized train/validation comparison stage.	Cache/temporary
src/models/step_04_robustness/01_time_series_robustness.py	.py	4.6 KB	source code	Time-series robustness stage.	Active or support
src/models/step_04_robustness/__init__.py	.py	36.0 B	source code	Time-series robustness stage.	Active or support
src/models/step_04_robustness/__pycache__/01_time_series_robustness.cpython-313.pyc	.pyc	5.7 KB	cache or temporary	Time-series robustness stage.	Cache/temporary
src/models/step_04_robustness/__pycache__/__init__.cpython-313.pyc	.pyc	182.0 B	cache or temporary	Time-series robustness stage.	Cache/temporary
src/models/step_05_test/01_final_test_evaluation.py	.py	13.3 KB	source code	One-time final test estimation and output stage.	Active or support
src/models/step_05_test/__init__.py	.py	24.0 B	source code	One-time final test estimation and output stage.	Active or support
src/models/step_05_test/__pycache__/01_final_test_evaluation.cpython-313.pyc	.pyc	6.2 KB	cache or temporary	One-time final test estimation and output stage.	Cache/temporary
src/models/step_05_test/__pycache__/__init__.cpython-313.pyc	.pyc	164.0 B	cache or temporary	One-time final test estimation and output stage.	Cache/temporary
src/models/step_06_interpretation/00_utils.py	.py	12.3 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/01_model_interpretation.py	.py	7.8 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/02_feature_importance.py	.py	6.4 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/03_create_result_tables.py	.py	13.3 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/04_create_figures.py	.py	12.0 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/05_report_notes.py	.py	14.1 KB	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/README.md	.md	2.1 KB	documentation	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/__init__.py	.py	61.0 B	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/step_06_interpretation/__pycache__/00_utils.cpython-313.pyc	.pyc	12.1 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary

Path	Type	Size	Class	Role	Status
src/models/step_06_interpretation/__pycache__/01_model_interpretation.cpython-313.pyc	.pyc	4.7 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/02_feature_importance.cpython-313.pyc	.pyc	9.1 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/03_create_result_tables.cpython-313.pyc	.pyc	9.6 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/04_create_figures.cpython-313.pyc	.pyc	10.2 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/05_report_notes.cpython-313.pyc	.pyc	9.2 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/__init__.cpython-313.pyc	.pyc	211.0 B	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/__pycache__/run_all.cpython-313.pyc	.pyc	1.0 KB	cache or temporary	Frozen-output interpretation and report-artifact stage.	Cache/temporary
src/models/step_06_interpretation/run_all.py	.py	569.0 B	source code	Frozen-output interpretation and report-artifact stage.	Active or support
src/models/utils/__init__.py	.py	50.0 B	source code	Shared model loading, estimation, or evaluation utility.	Active or support
src/models/utils/__pycache__/__init__.cpython-313.pyc	.pyc	183.0 B	cache or temporary	Shared model loading, estimation, or evaluation utility.	Cache/temporary
src/models/utils/__pycache__/data.cpython-313.pyc	.pyc	2.9 KB	cache or temporary	Shared model loading, estimation, or evaluation utility.	Cache/temporary
src/models/utils/__pycache__/estimation.cpython-313.pyc	.pyc	6.3 KB	cache or temporary	Shared model loading, estimation, or evaluation utility.	Cache/temporary
src/models/utils/__pycache__/evaluation.cpython-313.pyc	.pyc	4.2 KB	cache or temporary	Shared model loading, estimation, or evaluation utility.	Cache/temporary
src/models/utils/__pycache__/portfolio.cpython-313.pyc	.pyc	6.9 KB	cache or temporary	Shared model loading, estimation, or evaluation utility.	Cache/temporary
src/models/utils/data.py	.py	1.9 KB	source code	Shared model loading, estimation, or evaluation utility.	Active or support

Path	Type	Size	Class	Role	Status
src/models/utils/estimation.py	.py	4.4 KB	source code	Shared model loading, estimation, or evaluation utility.	Active or support
src/models/utils/evaluation.py	.py	2.9 KB	source code	Shared model loading, estimation, or evaluation utility.	Active or support
thesis/Makefile	Makefile	226.0 B	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/README.md	.md	1.7 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.aux	.aux	8.8 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.bbl	.bbl	0.0 B	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.bcf	.bcf	106.4 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.fdb_latexmk	.fdb_latexmk	25.7 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.fls	.fls	29.0 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.log	.log	53.7 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.out	.out	4.4 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.pdf	.pdf	1.1 MB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.run.xml	.xml	2.3 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/paper.tex	.tex	30.2 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review
thesis/references.bib	.bib	3.6 KB	documentation	Current course-paper source, build artifact, or bibliography.	Review

B.2 Producer, consumer, and reproducibility inventory

Path	Producer	Consumer	Essential	Reproducibility
.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
.gitignore	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
.matplotlib/fontlist-v390.json	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
README.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
backup/model_reorganization_20260719_054538/audit_report.txt	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/existing_output_files.txt	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/original_directory_tree.txt	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/01_fixed_linear_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/02_fixed_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/03_compare_fixed_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/01_optimized_linear_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/02_optimized_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/03_compare_optimized_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/gradient_boosting_inventory.txt	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/README.md	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/README_MODEL_WORKFLOW.md	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/02_fixed_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/03_compare_fixed_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/02_tune_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/03_compare_tuning_results.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/02_optimized_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/03_compare_optimized_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_04_robustness/01_time_series_robustness.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boosting_20260719_063836/src/models/step_05_test/01_final_test_evaluation.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/01_portfolio_ sorts.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/02_compare_po rtfolio_results.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/03_final_mode l_and_portfolio_summary.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
docs/Data_Construction_Guide.p df	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
docs/Data_Construction_Guide.t ex	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
docs/Pipeline_Overview.pdf	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
docs/figures/.gitkeep	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
input/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
input/README.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if referenced	External acquisition required
input/external/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
input/external/.gitkeep	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if referenced	External acquisition required
input/external/fama_french_rf_ monthly.csv	src/acquisition/02_create_fama_ french_rf_input.py	src/data/02_build_monthly_ stock_features.py	Yes if referenced	External acquisition required
input/external/welch_goyal_mac ro_1990_2025.csv	external preparation; validator 04_ create_welch_goyal_input.py	src/data/04_build_raw_kelly_ dataset.py	Yes if referenced	External acquisition required
input/fama_french_rf_monthly.c sv	src/acquisition/02_create_fama_ french_rf_input.py	src/data/02_build_monthly_ stock_features.py	Yes if referenced	External acquisition required
input/input_manifest.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if referenced	External acquisition required
input/market_gspc_daily.csv	src/acquisition/03_download_ market_inputs.py	src/data/02_build_monthly_ stock_features.py	Yes if referenced	External acquisition required
input/market_vix_daily.csv	src/acquisition/03_download_ market_inputs.py	src/data/02_build_monthly_ stock_features.py	Yes if referenced	External acquisition required
input/raw/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
input/raw/.gitkeep	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if referenced	External acquisition required
input/raw/compustat_annual_198 0_2025.csv	WRDS/Compustat extraction; not scripted	src/data/03_add_fundamentals_ and_macro.py	Yes if referenced	External acquisition required
input/stock_universe_locked.csv	src/acquisition/01_create_locked_ stock_universe.py	src/data/01_build_clean_yahoo_ daily.py	Yes if referenced	External acquisition required

Path	Producer	Consumer	Essential	Reproducibility
output/data/final/cleaning_summary.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/daily_prices_clean_1987_2026.csv	src/data/01_build_clean_yahoo_daily.py	src/data/02_build_monthly_stock_features.py	Yes if consumed	Reproducible only if inputs/code match
output/data/final/dropped_predictors_empty.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/extreme_target_counts.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/extreme_target_observations.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/kelly_raw_dataset_summary.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/model_dataset_kelly_ranked_full_1990_2025.parquet	src/data/05_clean_and_rank_normalize.py	src/models/utils/data.py	Yes if consumed	Reproducible only if inputs/code match
output/data/final/model_dataset_kelly_raw_full_1990_2025.csv	src/data/04_build_raw_kelly_dataset.py	src/data/05_clean_and_rank_normalize.py	Yes if consumed	Reproducible only if inputs/code match
output/data/final/monthly_imputation_medians_summary.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/monthly_stock_panel_with_targets_1990_2025.csv	src/data/02_build_monthly_stock_features.py	src/data/03_add_fundamentals_and_macro.py	Yes if consumed	Reproducible only if inputs/code match
output/data/final/predictor_columns_kelly_ranked.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/predictor_columns_kelly_raw.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/removed_tickers.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/final/sp500_tickers_clean.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/computat_annual_cleaned_1980_2025.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/daily_prices_raw_1987_2026.csv	src/data/01_build_clean_yahoo_daily.py	same script	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/monthly_panel_with_compustat_macro_1990_2025.csv	src/data/03_add_fundamentals_and_macro.py	src/data/04_build_raw_kelly_dataset.py	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/monthly_removed_tickers.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/monthly_return_quality_report.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/sp500_tickers_raw.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/data/intermediate/ticker_quality_report.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/decision_tree_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/fixed/diagnostics/decision_tree_rules.txt	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/decision_tree_split_feature_distributions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/decision_tree_structure.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/decision_tree_train_leaf_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/decision_tree_validation_leaf_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/diagnostic_warning_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/diagnostic_warnings.txt	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/fixed_data_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/fixed_model_performance_by_month.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/fixed_model_performance_by_validation_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/fixed_prediction_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/largest_absolute_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/largest_absolute_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/largest_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/largest_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/ols3_predictor_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/ols3_selected_predictors.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/prediction_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/fixed/diagnostics/predictor_distribution_flags.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/predictor_distribution_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/random_forest_extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/random_forest_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/random_forest_prediction_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/smallest_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/smallest_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/squared_error_concentration.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/target_extreme_counts.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/target_quantiles.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/target_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/target_summary_by_month.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/target_summary_by_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/diagnostics/validation_performance_by_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_all_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_line_ar_model_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_line_ar_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_line_ar_model_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_model_summary_wide.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_model_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_tree_model_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/fixed/fixed_tree_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/fixed/fixed_tree_model_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/best_hyperparameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/decision_tree_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/elastic_net_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/feature_importance_top_variables.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/prediction_distribution_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/prediction_vs_realized_pls.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/test_correlation_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/test_monthly_mse_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/test_oos_r2_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/top_decision_tree_features.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/top_elastic_net_coefficients_skipped.txt	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/top_pls_coefficients.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/top_random_forest_features.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/yearly_mse_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/figures/yearly_oos_r2_by_model.png	src/models/step_06_interpretation/04_create_figures.py	thesis/paper.tex	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/final_model_complexity.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/final_prediction_results.csv	src/models/step_06_interpretation/03_create_result_tables.py	thesis/paper.tex and interpretation figures	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/interpretation/final_report_results.xlsx	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/fixed_vs_optimized_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/model_behavior_flags.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/model_complexity_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/model_interpretation_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/ols_3_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/pls_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/pls_components.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/random_forest_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/report_notes.md	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/tree_model_complexity.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/interpretation/early_prediction_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_split_feature_distributions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_structure.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_train_leaf_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/decision_tree_validation_leaf_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/diagnostic_warning_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_data_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_model_performance_by_month.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_model_performance_by_validation_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/fixed_prediction_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/gradient_boosting_extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/gradient_boosting_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_absolute_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_absolute_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/largest_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/ols3_predictor_distribution.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/ols3_selected_predictors.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/prediction_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/predictor_distribution_flags.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/predictor_distribution_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/random_forest_prediction_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/smallest_prediction_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/smallest_target_rows.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/squared_error_concentration.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_extreme_counts.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_quantiles.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary_by_month.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/target_summary_by_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
output/models/legacy/gradient_boosting/mixed/fixed/diagnostics/validation_performance_by_year.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_all_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_model_summary_wide.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_model_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_tree_model_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_tree_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/fixed/fixed_tree_model_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_all_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_family_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/fixed_vs_optimized_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/optimized_all_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/optimized_model_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
output/models/legacy/gradient_boosting/mixed/optimization/optimized_tree_model_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/tuning/tuning_all_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/mixed/tuning/tuning_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/standalone/gradient_boosting_best_parameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/standalone/gradient_boosting_extreme_predictions.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/standalone/gradient_boosting_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/legacy/gradient_boosting/standalone/gradient_boosting_tuning_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	No for active run	Archival copy
output/models/optimization/fix_vs_optimized_all_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/fix_vs_optimized_family_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/fix_vs_optimized_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_all_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_linear_model_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_linear_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_linear_model_prediction.s.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_model_validation_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_tree_model_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/optimization/optimized_tree_model_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/optimization/optimized_tree_model_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/cumulative_long_short_returns.png	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/final_portfolio_report_table.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/final_prediction_portfolio_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/final_prediction_portfolio_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_annual_performance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_block_bootstrap.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_cumulative_returns.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_drawdown_statistics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_monthly_returns.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_ranking.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/portfolio/portfolio_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/robustness/time_series_robustness_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/robustness/time_series_robustness_predictions.parquet	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/robustness/time_series_robustness_yearly_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/test/final_test_metrics.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/test/final_test_model_comparison.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/test/final_test_predictions.parquet	src/models/step_05_test/01_final_test_evaluation.py	src/models/step_06_interpretation/*	Yes if consumed	Reproducible only if inputs/code match
output/models/test/interpretation_inputs/linear_model_coefficients.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/test/interpretation_inputs/model_complexity.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/test/interpretation_inputs/pls_components.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match

Path	Producer	Consumer	Essential	Reproducibility
output/models/test/interpretation_inputs/tree_feature_importance.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/decision_tree_best_parameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/decision_tree_tuning_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/elastic_net_best_parameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/elastic_net_tuning_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/pls_best_parameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/pls_tuning_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/random_forest_best_parameters.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/random_forest_tuning_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/tuning_all_results.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/models/tuning/tuning_summary.csv	Model-stage script matching directory	Later comparison/interpretation scripts	Yes if consumed	Reproducible only if inputs/code match
output/reports/kelly_characteristic_comparison.csv	Not verifiable from the current repository.	Not verifiable from the current repository.	Yes if consumed	Reproducible only if inputs/code match
requirements.txt	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
run_pipeline.sh	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/__pycache__/config.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/__pycache__/feature_definitions.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/01_create_locked_stock_universe.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/acquisition/02_create_fama_french_rf_input.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/acquisition/03_download_market_inputs.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/acquisition/04_create_welch_goyal_input.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/acquisition/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic

Path	Producer	Consumer	Essential	Reproducibility
src/acquisition/__pycache__/01_create_locked_stock_universe.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/__pycache__/02_create_fama_french_rf_input.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/__pycache__/03_download_market_inputs.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/__pycache__/04_create_welch_goyal_input.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/__pycache__/manifest.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/acquisition/manifest.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/config.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/01_build_clean_yahoo_daily.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/02_build_monthly_stock_features.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/03_add_fundamentals_and_macro.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/04_build_raw_kelly_dataset.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/05_clean_and_rank_normalize.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/data/__pycache__/01_build_clean_yahoo_daily.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/__pycache__/02_build_monthly_stock_features.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/__pycache__/03_add_fundamentals_and_macro.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/__pycache__/04_build_raw_kelly_dataset.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/__pycache__/05_clean_and_rank_normalize.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable

Path	Producer	Consumer	Essential	Reproducibility
src/data/__pycache__/06_inspect_final_dataset.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/data/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/documentation/latex/Pipeline_Overview.aux	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.fdb_latexmk	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.fls	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.log	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.out	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.pdf	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.synctex.gz	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/documentation/latex/Pipeline_Overview.tex	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/feature_definitions.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/.DS_Store	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/README.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/README_MODEL_WORKFLOW.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/legacy/gradient_boosting/source_snapshots/README.md	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/README_MODEL_WORKFLOW.md	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/step_01_fixed/02_fixed_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/02_tune_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/03_compare_tuning_results.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy

Path	Producer	Consumer	Essential	Reproducibility
src/models/legacy/gradient_boosting/source_snapshots/step_03_optimization/02_optimized_tree_models.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/step_04_robustness/01_time_series_robustness.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/legacy/gradient_boosting/source_snapshots/step_05_test/01_final_test_evaluation.py	Historical reorganization/removal process	No active consumer found	No for active run	Archival copy
src/models/step_01_fixed/01_fixed_linear_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_01_fixed/02_fixed_tree_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_01_fixed/03_compare_fixed_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_01_fixed/04_diagnose_fixed_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_01_fixed/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_01_fixed/__pycache__/01_fixed_linear_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_01_fixed/__pycache__/02_fixed_tree_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_01_fixed/__pycache__/03_compare_fixed_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_01_fixed/__pycache__/04_diagnose_fixed_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_01_fixed/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_02_tuning/01_tune_linear_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_02_tuning/02_tune_tree_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_02_tuning/03_compare_tuning_results.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_02_tuning/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_02_tuning/__pycache__/01_tune_linear_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable

Path	Producer	Consumer	Essential	Reproducibility
src/models/step_02_tuning/__pycache__/02_tune_tree_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_02_tuning/__pycache__/03_compare_scaling.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_02_tuning/__pycache__/03_compare_tuning_results.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_02_tuning/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/01_optimized_linear_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_03_optimization/02_optimized_tree_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_03_optimization/03_compare_optimized_models.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_03_optimization/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_03_optimization/__pycache__/01_optimized_linear_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/02_optimized_tree_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/03_compare_optimized_models.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/04_compare_fixed_vs_optimized.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/05_time_series_robustness.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/06_final_test_evaluation.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_03_optimization/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_04_robustness/01_time_series_robustness.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_04_robustness/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic

Path	Producer	Consumer	Essential	Reproducibility
src/models/step_04_robustness/____pycache__/01_time_series_robustness.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_04_robustness/____pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_05_test/01_final_test_evaluation.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_05_test/____init____.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_05_test/____pycache__/01_final_test_evaluation.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_05_test/____pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/00_utils.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/01_model_interpretation.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/02_feature_importance.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/03_create_result_tables.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/04_create_figures.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/05_report_notes.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/README.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/____init____.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/step_06_interpretation/____pycache__/00_utils.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/____pycache__/01_model_interpretation.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/____pycache__/02_feature_importance.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/____pycache__/03_create_result_tables.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/____pycache__/04_create_figures.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/____pycache__/05_report_notes.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable

Path	Producer	Consumer	Essential	Reproducibility
src/models/step_06_interpretation/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/__pycache__/run_all.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/step_06_interpretation/run_all.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/utils/__init__.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/utils/__pycache__/__init__.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/utils/__pycache__/data.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/utils/__pycache__/estimation.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/utils/__pycache__/evaluation.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/utils/__pycache__/portfolio.cpython-313.pyc	Not verifiable from the current repository.	Not verifiable from the current repository.	No	Regenerable
src/models/utils/data.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/utils/estimation.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
src/models/utils/evaluation.py	Not verifiable from the current repository.	Not verifiable from the current repository.	Usually	Source-controlled logic
thesis/Makefile	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/README.md	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.aux	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.bbl	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.bcf	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.fdb_latexmk	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.fls	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.log	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.out	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.pdf	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/paper.run.xml	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented

Path	Producer	Consumer	Essential	Reproducibility
thesis/paper.tex	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented
thesis/references.bib	Not verifiable from the current repository.	Not verifiable from the current repository.	Context dependent	Unclear or documented

C Complete File-by-File and Function Documentation

C.1 File 1: .gitignore

Identity. .gitignore. **Role.** Repository support file; exact role follows from its category and name. Type .gitignore, size 632.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.2 File 2: README.md

Identity. README.md. **Role.** Repository support file; exact role follows from its category and name. Type .md, size 2.4 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.3 File 3: audit_report.txt

Identity. backup/model_reorganization_20260719_054538/audit_report.txt. **Role.** Archived snapshot; not part of the active workflow. Type .txt, size 6.0 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.4 File 4: existing_output_files.txt

Identity. backup/model_reorganization_20260719_054538/existing_output_files.txt. **Role.** Archived snapshot; not part of the active workflow. Type .txt, size 3.9 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.5 File 5: original_directory_tree.txt

Identity. backup/model_reorganization_20260719_054538/original_directory_tree.txt.

Role. Archived snapshot; not part of the active workflow. Type .txt, size 902.0 B. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.6 File 6: 01_fixed_linear_models.py

Identity. backup/model_reorganization_20260719_054538/src/models/step_01_fixed/01_fixed_linear_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 3.3 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 139 lines; direct main block: yes; imports: numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.evaluation:evaluate_model, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fixed_models</code>	<code>all_predictors</code>	Define the fixed linear model specifications.	See docstring; no additional side effect verified.	28
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	63

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.7 File 7: 02 fixed tree models.py

Identity. backup/model_reorganization_20260719_054538/src/models/step_01_fixed/02_fixed_tree_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 4.2 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 157 lines; direct main block: yes; imports: pandas, pathlib:Path, sklearn.ensemble: GradientBoostingRegressor, RandomForestRegressor, sklearn.tree:DecisionTreeRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation: evaluate_model, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	none	Define the optimized tree-model specifications.	See docstring; no additional side effect verified.	24
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	82

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.8 File 8: 03 compare fixed models.py

Identity. backup/model_reorganization_20260719_054538/src/models/step_01_fixed/03_compare_fixed_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 1.2 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 61 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	18

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.9 File 9: 01 optimized linear models.py

Identity. backup/model_reorganization_20260719_054538/src/models/step_03_optimization/01_optimized_linear_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 3.9 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 156 lines; direct main block: yes; imports: numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_model, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	<code>all_predictors</code>	Define the optimized linear model specifications.	See docstring; no additional side effect verified.	30
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	80

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.10 File 10: 02 optimized tree models.py

Identity. `backup/model_reorganization_20260719_054538/src/models/step_03_optimization/02_optimized_tree_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 2.7 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 76 lines; direct main block: yes; imports: `pathlib:Path`, `sklearn.ensemble:GradientBoostingRegressor`, `RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:load_model_data`, `src.models.utils.estimation:fit_models`, `load_best_parameters`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	<code>parameters</code>	Construct tree models from validation-selected parameters.	See docstring; no additional side effect verified.	21
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	52

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.11 File 11: 03 compare optimized models.py

Identity. `backup/model_reorganization_20260719_054538/src/models/step_03_optimization/03_compare_optimized_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 1.5 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 70 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	18

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.12 File 12: 04 compare fixed vs optimized.py

Identity. backup/model_reorganization_20260719_054538/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 5.0 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 197 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	19

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.13 File 13: estimation.py

Identity. backup/model_reorganization_20260719_054538/src/models/utils/estimation.py.

Role. Archived snapshot; not part of the active workflow. Type .py, size 3.6 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 141 lines; direct main block: no; imports: ast:literal_eval, numpy, pandas, sklearn.model_selection:ParameterGrid, src.config:TARGET, src.models.utils.data:arrays, src.models.utils.evaluation:evaluate_model, monthly_mse.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fit_models</code>	<code>models, samples, predictors, target</code>	Fit models on development data and evaluate their predictions.	See docstring; no additional side effect verified.	16
<code>tune_grid</code>	<code>family, grid, make_model, data, predictors, folds, target</code>	Test all hyperparameter combinations and select the lowest CV MSE.	See docstring; no additional side effect verified.	53
<code>save_results</code>	<code>family, results, best_params, results_file, parameters_file</code>	Save tuning results and the best parameters.	See docstring; no additional side effect verified.	120
<code>load_best_parameters</code>	<code>path</code>	Load the saved best parameter dictionary.	See docstring; no additional side effect verified.	138

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.14 File 14: `evaluation.py`

Identity. `backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py`.

Role. Archived snapshot; not part of the active workflow. Type `.py`, size 2.9 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 106 lines; direct main block: no; imports: `numpy`, `pandas`, `sklearn.metrics:mean_absolute_error,mean_squared_error`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>monthly_mse</code>	<code>y_true, y_pred, months</code>	Calculate MSE for each month and average equally across months.	See docstring; no additional side effect verified.	8
<code>evaluate_predictions</code>	<code>y_true, y_pred, benchmark, months</code>	Calculate prediction metrics.	See docstring; no additional side effect verified.	18
<code>evaluate_model</code>	<code>model_name, samples, predictions, target, benchmark_mean</code>	Evaluate one model and create its prediction table.	See docstring; no additional side effect verified.	51
<code>rank_models</code>	<code>metrics, sample</code>	Rank models by monthly MSE.	See docstring; no additional side effect verified.	96

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.15 File 15: gradient boosting inventory.txt

Identity. backup/remove_gradient_boosting_20260719_063836/gradient_boosting_inventory.txt. **Role.** Archived snapshot; not part of the active workflow. Type .txt, size 3.1 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.16 File 16: README.md

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/README.md. **Role.** Archived snapshot; not part of the active workflow. Type .md, size 1.7 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.17 File 17: README MODEL WORKFLOW.md

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/README_MODEL_WORKFLOW.md. **Role.** Archived snapshot; not part of the active workflow. Type .md, size 2.8 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.18 File 18: 02 fixed tree models.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/02_fixed_tree_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 2.9 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 94 lines; direct main block: yes; imports: pandas, pathlib:Path, sklearn.ensemble: GradientBoostingRegressor, RandomForestRegressor, sklearn.tree:DecisionTreeRegressor,

```
src.config:MODEL_OUTPUT_DIR,TARGET, src.models.utils.data:arrays,load_model_data,  
src.models.utils.evaluation:evaluate_model, sys.
```

Function	Inputs	Return/output meaning	Side effects	Line
<code>fixed_models</code>	none	Define the fixed tree-model specifications.	See docstring; no additional side effect verified.	21
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	46

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.19 File 19: 03 compare fixed models.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/03_compare_fixed_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 1.4 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 64 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	18

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.20 File 20: 02 tune tree models.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/02_tune_tree_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 3.1 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 140 lines; direct main block: yes; imports: `pathlib:Path`, `sklearn.ensemble:GradientBoostingRegressor`, `RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:expanding_year_folds`, `load_model_data`, `src.models.utils.estimation:save_results`, `tune_grid`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>make_model</code>	<code>family, params</code>	Construct one tree-model candidate.	See docstring; no additional side effect verified.	74
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	96

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.21 File 21: 03 compare tuning results.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/03_compare_tuning_results.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 1.3 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 51 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	23

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.22 File 22: 02 optimized tree models.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/02_optimized_tree_models.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 3.8 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 108 lines; direct main block: yes; imports: pandas, pathlib:Path, sklearn.ensemble: GradientBoostingRegressor, RandomForestRegressor, sklearn.tree:DecisionTreeRegressor, src.config:MODEL_OUTPUT_DIR,TARGET, src.models.utils.data:arrays,load_model_data, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation: evaluate_model, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	none	Define tree models from the saved tuning results.	See docstring; no additional side effect verified.	23
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	60

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.23 File 23: 03 compare optimized models.py

Identity. `backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/03_compare_optimized_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 1.7 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 73 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	18

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.24 File 24: 04 compare fixed vs optimized.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 5.6 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 214 lines; direct main block: yes; imports: pandas, pathlib:Path, src.config:MODEL_OUTPUT_DIR, src.models.utils.evaluation:rank_models, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	19

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.25 File 25: 01 time series robustness.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_04_robustness/01_time_series_robustness.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 4.2 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 123 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.ensemble:GradientBoostingRegressor, RandomForestRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:load_model_data, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_predictions, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_factories</code>	<code>none</code>	Return fresh optimized tree models for each annual refit.	See docstring; no additional side effect verified.	25
<code>walk_forward</code>	<code>data, predictors, model_name, make_model, window</code>	Generate annual validation forecasts using past observations only.	See docstring; no additional side effect verified.	56
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	86

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.26 File 26: 01 final test evaluation.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_05_test/01_final_test_evaluation.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 7.0 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 212 lines; direct main block: yes; imports: numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.ensemble:GradientBoostingRegressor, RandomForestRegressor, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, sklearn.tree:DecisionTreeRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_model, rank_models, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_specifications</code>	<code>predictors</code>	Return the prespecified fixed and optimized models.	See docstring; no additional side effect verified.	28
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	144

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.27 File 27: 01 portfolio sorts.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/01_portfolio_sorts.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 3.1 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 90 lines; direct main block: yes; imports: pandas, pathlib:Path, src.config:MODEL_OUTPUT_DIR,TARGET, src.models.utils.portfolio:block_bootstrap_mean,evaluate_portfolio,rank_portfolios, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>prediction_groups</code>	none	Yield final-test model predictions.	See docstring; no additional side effect verified.	21
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	48

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.28 File 28: 02 compare portfolio results.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/02_compare_portfolio_results.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 1.8 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 57 lines; direct main block: yes; imports: pandas, pathlib:Path, src.config:MODEL_OUTPUT_DIR, src.models.utils.portfolio:rank_portfolios, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>select_columns</code>	<code>data</code>	No function docstring.	See docstring; no additional side effect verified.	26
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	30

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.29 File 29: 03 final model and portfolio summary.py

Identity. backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/03_final_model_and_portfolio_summary.py. **Role.** Archived snapshot; not part of the active workflow. Type .py, size 1.5 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 46 lines; direct main block: yes; imports: pandas, pathlib:Path, src.config:MODEL_OUTPUT_DIR, src.models.utils.portfolio:rank_portfolios, sys.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	18

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.30 File 30: Data Construction Guide.tex

Identity. docs/Data_Construction_Guide.tex. **Role.** Earlier data-construction or pipeline documentation. Type .tex, size 23.8 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.31 File 31: README.md

Identity. input/README.md. **Role.** Permanent local input or input metadata. Type .md, size 594.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Permanent local input. External acquisition provenance varies; Compustat extraction is not scripted.

C.32 File 32: decision tree rules.txt

Identity. output/models/fixed/diagnostics/decision_tree_rules.txt. **Role.** Frozen model, diagnostic, portfolio, or interpretation output. Type .txt, size 271.0 B. Producer/input relationship: Model-stage script matching directory. Consumer/dependent relationship: Later comparison/interpretation scripts.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Frozen evidence. Reproducibility depends on matching code/input versions; no provenance hashes are stored.

C.33 File 33: diagnostic warnings.txt

Identity. output/models/fixed/diagnostics/diagnostic_warnings.txt. **Role.** Frozen model, diagnostic, portfolio, or interpretation output. Type .txt, size 227.0 B. Producer/input relationship: Model-stage script matching directory. Consumer/dependent relationship: Later comparison/interpretation scripts.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Frozen evidence. Reproducibility depends on matching code/input versions; no provenance hashes are stored.

C.34 File 34: top elastic net coefficients skipped.txt

Identity. output/models/interpretation/figures/top_elastic_net_coefficients_skipped.txt. **Role.** Frozen model, diagnostic, portfolio, or interpretation output. Type .txt, size 189.0 B. Producer/input relationship: src/models/step_06_interpretation/04_create_figures.py. Consumer/dependent relationship: thesis/paper.tex.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Frozen evidence. Reproducibility depends on matching code/input versions; no provenance hashes are stored.

C.35 File 35: report notes.md

Identity. output/models/interpretation/report_notes.md. **Role.** Frozen model, diagnostic, portfolio, or interpretation output. Type .md, size 7.0 KB. Producer/input relationship: Model-stage script matching directory. Consumer/dependent relationship: Later comparison/interpretation scripts.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Frozen evidence. Reproducibility depends on matching code/input versions; no provenance hashes are stored.

C.36 File 36: requirements.txt

Identity. requirements.txt. **Role.** Unpinned Python dependency list. Type .txt, size 84.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.37 File 37: run pipeline.sh

Identity. run_pipeline.sh. **Role.** Intended end-to-end execution wrapper; currently stale after model reorganization. Type .sh, size 1.4 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Uses fail-fast shell options but references five obsolete paths, so it cannot currently complete.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.38 File 38: init .py

Identity. `src/__init__.py`. **Role.** Repository support file; exact role follows from its category and name. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.39 File 39: 01 create locked stock universe.py

Identity. `src/acquisition/01_create_locked_stock_universe.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 3.5 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 128 lines; direct main block: yes; imports: `datetime:date`, `pandas`, `pathlib:Path`, `requests`, `src.acquisition.manifest:update_input_manifest`, `src.config:START_YEAR`, `END_YEAR`, `WIKI_URL`, `INPUT_MANIFEST_FILE`, `STOCK_UNIVERSE_FILE`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>clean_ticker</code>	<code>value</code>	Convert ticker to Yahoo format, for example BRK.B -> BRK-B.	See docstring; no additional side effect verified.	28
<code>normalize_columns</code>	<code>table</code>	Return a copy with simple string column names.	See docstring; no additional side effect verified.	41
<code>find_column</code>	<code>table, words</code>	Find the first column containing all requested words.	See docstring; no additional side effect verified.	55
<code>build_sp500_universe</code>	<code>none</code>	Combine current and historical Wikipedia S&P 500 tickers.	See docstring; no additional side effect verified.	64
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	108

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.40 File 40: 02 create fama french rf input.py

Identity. `src/acquisition/02_create_fama_french_rf_input.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 2.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 75 lines; direct main block: yes; imports: `io:BytesIO`, `pandas`, `pathlib:Path`, `requests`, `src.acquisition.manifest:update_input_manifest`, `src.config:FF_URL`, `FAMA_FRENCH_RF_FILE`, `INPUT_MANIFEST_FILE`, `sys`, `zipfile:ZipFile`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>download_fama_french_factors</code>	none	Download and parse monthly Fama-French three-factor data.	See docstring; no additional side effect verified.	19
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	52

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.41 File 41: 03 download market inputs.py

Identity. `src/acquisition/03_download_market_inputs.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 2.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 81 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.acquisition.manifest:update_input_manifest`, `src.config:PRICE_START,PRICE_END,GSPC_DAILY_FILE,VIX_DAILY_FILE,INPUT_MANIFEST_FILE`, `sys`, `yfinance`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>download_yahoo_daily</code>	<code>ticker</code>	Download one Yahoo daily price series with standard columns.	See docstring; no additional side effect verified.	26
<code>save_market_input</code>	<code>ticker</code> , <code>output_file</code> , <code>source_name</code>	Download, save, and register one market input.	See docstring; no additional side effect verified.	56
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	75

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.42 File 42: 04 create welch goyal input.py

Identity. `src/acquisition/04_create_welch_goyal_input.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 2.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 66 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.acquisition.manifest:update_input_manifest`, `src.config:INPUT_MANIFEST_FILE`, `WELCH_GOYAL_INPUT_FILE`, `src.feature_definitions:LOCKED_MACRO_COLUMNS`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
load_and_validate_welch_goyal	none	Validate the local cleaned Welch-Goyal monthly input.	See docstring; no additional side effect verified.	22
main	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	49

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.43 File 43: `init.py`

Identity. `src/acquisition/__init__.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 48.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 1 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.44 File 44: `manifest.py`

Identity. `src/acquisition/manifest.py`. **Role.** One-time acquisition or input-registration logic. Type `.py`, size 1.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 44 lines; direct main block: no; imports: `datetime:date`, `pandas`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>update_input_manifest</code>	<code>manifest_file</code> , <code>input_file</code> , <code>source</code> , <code>coverage_start</code> , <code>coverage_end</code> , <code>notes</code>	Insert or replace one manifest row for a permanent input file.	See docstring; no additional side effect verified.	18

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.45 File 45: config.py

Identity. src/config.py. **Role.** Central paths, dates, target, and sample boundaries. Type .py, size 4.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 106 lines; direct main block: no; imports: pandas, pathlib:Path.

No functions are defined.

Critical decisions. Centralizes 1990-2025 sample, 1987-2026 price window, target name, train end 2014, validation end 2019, and all canonical paths. Importing it creates directories as a side effect.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.46 File 46: 01 build clean yahoo daily.py

Identity. src/data/01_build_clean_yahoo_daily.py. **Role.** Active data-construction stage. Type .py, size 7.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 259 lines; direct main block: yes; imports: pandas, pathlib:Path, src.config:PRICE_START,PRICE_END,SLEEP_SECONDS,DAILY_BAD_ROW_SHARE_THRESHOLD,STOCK_UNIVERSE_FILE,TICKERS_RAW_FILE,DAILY_RAW_FILE,QUALITY_REPORT_FILE,TICKERS_CLEAN_FILE,DAILY_CLEAN_FILE,REMOVED_TICKERS_FILE, sys, time, yfinance.

Function	Inputs	Return/output meaning	Side effects	Line
<code>load_locked_universe</code>	<code>none</code>	Load unique Yahoo-format tickers.	See docstring; no additional side effect verified.	32
<code>download_daily_prices</code>	<code>tickers</code>	Download prices and return the data and failed ticker list.	See docstring; no additional side effect verified.	53
<code>clean_daily_prices</code>	<code>daily</code>	Remove bad rows and ticker histories with excessive bad data.	See docstring; no additional side effect verified.	126
<code>main</code>	<code>none</code>	Download, clean, and save the daily stock dataset.	Writes/fits only if stated by the function or enclosing main workflow.	217

Critical decisions. Uses a locked universe; downloads each ticker serially; flags OHLC errors, missing core fields, nonpositive prices, extreme returns, and duplicate dates; removes rows and tickers above a 10 percent bad-row share.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.47 File 47: 02 build monthly stock features.py

Identity. `src/data/02_build_monthly_stock_features.py`. **Role.** Active data-construction stage. Type `.py`, size 12.1 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 442 lines; direct main block: yes; imports: `numpy`, `pandas`, `pathlib:Path`, `src.config`: `SAMPLE_START`, `SAMPLE_END`, `TARGET`, `DAILY_CLEAN_FILE`, `GSPC_DAILY_FILE`, `VIX_DAILY_FILE`, `FAM_A_FRENCH_RF_FILE`, `MONTHLY_QUALITY_REPORT_FILE`, `MONTHLY_REMOVED_TICKERS_FILE`, `MONTHLY_STOCK_FILE`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>build_monthly_panel</code>	<code>daily</code>	Aggregate daily prices into monthly stock characteristics.	See docstring; no additional side effect verified.	30
<code>remove_bad_monthly_tickers</code>	<code>monthly</code>	Remove tickers with repeated bad returns and drop isolated bad months.	See docstring; no additional side effect verified.	85
<code>compound</code>	<code>returns</code>	Compound simple returns.	See docstring; no additional side effect verified.	147
<code>add_stock_features</code>	<code>monthly</code>	Create momentum, volatility, liquidity, and trend predictors.	See docstring; no additional side effect verified.	152
<code>load_market_data</code>	<code>none</code>	Load permanent daily S&P 500 and VIX files.	See docstring; no additional side effect verified.	209
<code>add_market_features</code>	<code>monthly, market</code>	Add monthly market return, market volatility, and VIX variables.	See docstring; no additional side effect verified.	235
<code>add_beta_idiovol</code>	<code>monthly, daily, market</code>	Create rolling market beta and idiosyncratic volatility.	See docstring; no additional side effect verified.	286
<code>load_fama_french_rf</code>	<code>none</code>	Load the permanent monthly risk-free rate.	See docstring; no additional side effect verified.	347
<code>add_target</code>	<code>monthly, rf</code>	Create next-month stock and excess returns.	See docstring; no additional side effect verified.	365
<code>main</code>	<code>none</code>	Build and save the monthly stock panel.	Writes/fits only if stated by the function or enclosing main workflow.	393

Critical decisions. Uses month-end adjusted-close returns, full rolling windows, daily 252/126 risk windows, current-month market/VIX, shifted next-observation return, and shifted RF.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.48 File 48: 03 add fundamentals and macro.py

Identity. `src/data/03_add_fundamentals_and_macro.py`. **Role.** Active data-construction stage. Type `.py`, size 11.1 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 428 lines; direct main block: yes; imports: `numpy`, `pandas`, `pathlib:Path`, `src.config:TARGET`, `MONTHLY_STOCK_FILE`, `COMPUSTAT_RAW_FILE`, `COMPUSTAT_CLEAN_FILE`, `PANEL_WITH_FUNDAMENTALS_FILE`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>column</code>	<code>data, name</code>	Return a column or a missing-value series.	See docstring; no additional side effect verified.	25
<code>ratio</code>	<code>numerator, denominator</code>	Calculate a ratio safely.	See docstring; no additional side effect verified.	33
<code>clean_compustat</code>	<code>none</code>	Create compact annual accounting predictors with a six-month lag.	See docstring; no additional side effect verified.	50
<code>merge_compustat</code>	<code>monthly, comp</code>	Attach the latest available annual Compustat report.	See docstring; no additional side effect verified.	374
<code>main</code>	<code>none</code>	Merge annual Compustat fundamentals into the monthly panel.	Writes/fits only if stated by the function or enclosing main workflow.	408

Critical decisions. Filters standard consolidated industrial USD records, builds ratios with zero-denominator protection, delays annual data six months, and uses a backward ticker-level as-of merge.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.49 File 49: 04 build raw kelly dataset.py

Identity. `src/data/04_build_raw_kelly_dataset.py`. **Role.** Active data-construction stage. Type `.py`, size 5.4 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 259 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:TARGET`, `PANEL_WITH_FUNDAMENTALS_FILE`, `WELCH_GOYAL_INPUT_FILE`, `RAW_KELLY_FILE`, `RAW_PREDICTOR_FILE`, `RAW_KELLY_SUMMARY_FILE`, `src.feature_definitions:LOCKED_CHARACTERISTIC_COLUMNS`, `LOCKED_MARKET_COLUMNS`, `LOCKED_MACRO_COLUMNS`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>add_industry_dummies</code>	<code>data</code>	Create SIC2 industry dummies.	See docstring; no additional side effect verified.	37
<code>add_welch_goyal_macro</code>	<code>data</code>	Merge monthly Welch-Goyal macro variables.	See docstring; no additional side effect verified.	62
<code>validate_columns</code>	<code>data, columns, label</code>	Confirm that the required variables are available.	See docstring; no additional side effect verified.	87
<code>prepare_raw_data</code>	<code>data, characteristics, market_variables, macro_variables, sic2_dummies</code>	Select and save the raw base predictor panel.	See docstring; no additional side effect verified.	103
<code>main</code>	<code>none</code>	Build and save the raw Kelly-style base dataset.	Writes/fits only if stated by the function or enclosing main workflow.	173

Critical decisions. Creates SIC2 dummies, merges same-month Welch-Goyal variables, locks 47 characteristics, four market variables, and eight macro variables.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.50 File 50: 05 clean and rank normalize.py

Identity. `src/data/05_clean_and_rank_normalize.py`. **Role.** Active data-construction stage. Type `.py`, size 10.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 474 lines; direct main block: yes; imports: `numpy`, `pandas`, `pathlib:Path`, `src.config:TARGET`, `RAW_KELLY_FILE`, `RAW_PREDICTOR_FILE`, `CLEAN_FULL_FILE`, `CLEAN_PREDICTOR_FILE`, `CLEANING_SUMMARY_FILE`, `DROPPED_PREDICTORS_FILE`, `MONTHLY_MEDIAN_FILE`, `EXTREME_TARGET_FILE`, `EXTREME_TARGET_COUNT_FILE`, `src.feature_definitions:LOCKED_CHARACTERISTIC_COLUMNS`, `LOCKED_MARKET_COLUMNS`, `LOCKED_MACRO_COLUMNS`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>target_diagnostics</code>	<code>data</code>	Document unusually large target returns.	See docstring; no additional side effect verified.	37
<code>impute_predictors</code>	<code>data, predictors</code>	Impute missing predictors using monthly cross-sectional medians.	See docstring; no additional side effect verified.	71
<code>rank_characteristics</code>	<code>data, characteristics</code>	Map monthly characteristic ranks to the interval [-1, 1].	See docstring; no additional side effect verified.	108
<code>create_interactions</code>	<code>data, characteristics, macro_variables</code>	Create ranked-characteristic and macro interactions.	See docstring; no additional side effect verified.	134
<code>check_final_data</code>	<code>data, predictors, characteristics</code>	Check the final model-ready dataset.	See docstring; no additional side effect verified.	169
<code>main</code>	<code>none</code>	Create and save the final model-ready dataset.	Writes/fits only if stated by the function or enclosing main workflow.	210

Critical decisions. Drops missing targets, median-imputes within month, zero-fills all-missing monthly predictors, ranks 47 characteristics to $[-1,1]$, creates 376 interactions, and leaves target unwinsorized.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.51 File 51: `init.py`

Identity. `src/data/__init__.py`. **Role.** Active data-construction stage. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Generally explicit and chronological. Main risks are next-observation target gaps, same-month macro timing, vendor/ticker quality, and large CSV memory use.

C.52 File 52: `Pipeline Overview.tex`

Identity. `src/documentation/latex/Pipeline_Overview.tex`. **Role.** Earlier data-construction or pipeline documentation. Type `.tex`, size 4.7 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.53 File 53: `feature_definitions.py`

Identity. `src/feature_definitions.py`. **Role.** Locked predictor-family definitions. Type `.py`, size 1.6 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 92 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.54 File 54: `README.md`

Identity. `src/models/README.md`. **Role.** Repository support file; exact role follows from its category and name. Type `.md`, size 1.8 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.55 File 55: README_MODEL_WORKFLOW.md

Identity. `src/models/README_MODEL_WORKFLOW.md`. **Role.** Repository support file; exact role follows from its category and name. Type `.md`, size 3.0 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.56 File 56: `init.py`

Identity. `src/models/__init__.py`. **Role.** Repository support file; exact role follows from its category and name. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.57 File 57: README.md

Identity. `src/models/legacy/gradient_boosting/source_snapshots/README.md`. **Role.** Archived snapshot; not part of the active workflow. Type `.md`, size 1.7 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.58 File 58: README_MODEL_WORKFLOW.md

Identity. `src/models/legacy/gradient_boosting/source_snapshots/README_MODEL_WORKFLOW.md`. **Role.** Archived snapshot; not part of the active workflow. Type `.md`, size 2.8 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.59 File 59: 02 fixed tree models.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_01_fixed/02_fixed_tree_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 2.9 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 94 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `sklearn.ensemble:GradientBoostingRegressor, RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR, TARGET`, `src.models.utils.data:arrays, load_model_data`, `src.models.utils.evaluation:evaluate_model`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fixed_models</code>	none	Define the fixed tree-model specifications.	See docstring; no additional side effect verified.	21
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	46

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.60 File 60: 02 tune tree models.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/02_tune_tree_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 3.1 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 140 lines; direct main block: yes; imports: `pathlib:Path`, `sklearn.ensemble:GradientBoostingRegressor`, `RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:expanding_year_folds`, `load_model_data`, `src.models.utils.estimation:save_results`, `tune_grid`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>make_model</code>	<code>family, params</code>	Construct one tree-model candidate.	See docstring; no additional side effect verified.	74
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	96

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.61 File 61: 03 compare tuning results.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/03_compare_tuning_results.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 1.3 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 51 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	23

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.62 File 62: 02 optimized tree models.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_03_optimization/02_optimized_tree_models.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 3.8 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 108 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `sklearn.ensemble:GradientBoostingRegressor`, `RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:arrays`, `load_model_data`, `src.models.utils.estimation:load_best_parameters`, `src.models.utils.evaluation:evaluate_model`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	none	Define tree models from the saved tuning results.	See docstring; no additional side effect verified.	23
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	60

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.63 File 63: 01 time series robustness.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_04_robustness/01_time_series_robustness.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 4.2 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 123 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.ensemble:GradientBoostingRegressor, RandomForestRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:load_model_data, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_predictions, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_factories</code>	<code>none</code>	Return fresh optimized tree models for each annual refit.	See docstring; no additional side effect verified.	25
<code>walk_forward</code>	<code>data, predictors, model_name, make_model, window</code>	Generate annual validation forecasts using past observations only.	See docstring; no additional side effect verified.	56
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	86

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.64 File 64: 01 final test evaluation.py

Identity. `src/models/legacy/gradient_boosting/source_snapshots/step_05_test/01_final_test_evaluation.py`. **Role.** Archived snapshot; not part of the active workflow. Type `.py`, size 7.0 KB. Producer/input relationship: Historical reorganization/removal process. Consumer/dependent relationship: No active consumer found.

Execution. 212 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.ensemble:GradientBoostingRegressor, RandomForestRegressor, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, sklearn.tree:DecisionTreeRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_model, rank_models, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_specifications</code>	<code>predictors</code>	Return the prespecified fixed and optimized models.	See docstring; no additional side effect verified.	28
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	144

Critical decisions. Archival decisions may differ from the active workflow and must not be used to interpret current frozen results.

Quality assessment. Clear archival value, but high stale-output and duplicated-logic risk if confused with the active pipeline.

C.65 File 65: 01 fixed linear models.py

Identity. `src/models/step_01_fixed/01_fixed_linear_models.py`. **Role.** Fixed train/validation benchmark stage. Type `.py`, size 3.9 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 159 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.evaluation:evaluate_model, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fixed_models</code>	<code>all_predictors</code>	Define the fixed linear model specifications.	See docstring; no additional side effect verified.	28
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	63

Critical decisions. Fits fixed specifications on train and reports train/validation metrics; test rows are not used.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.66 File 66: 02 fixed tree models.py

Identity. `src/models/step_01_fixed/02_fixed_tree_models.py`. **Role.** Fixed train/validation benchmark stage. Type `.py`, size 2.7 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 89 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `sklearn.ensemble:RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR, TARGET`, `src.models.utils.data:arrays, load_model_data`, `src.models.utils.evaluation:evaluate_model`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fixed_models</code>	none	Define the fixed tree-model specifications.	See docstring; no additional side effect verified.	21
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	41

Critical decisions. Fits fixed specifications on train and reports train/validation metrics; test rows are not used.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.67 File 67: 03 compare fixed models.py

Identity. `src/models/step_01_fixed/03_compare_fixed_models.py`. **Role.** Fixed train/validation benchmark stage. Type `.py`, size 1.6 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 73 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	26

Critical decisions. Fits fixed specifications on train and reports train/validation metrics; test rows are not used.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.68 File 68: 04 diagnose fixed models.py

Identity. `src/models/step_01_fixed/04_diagnose_fixed_models.py`. **Role.** Fixed train/validation benchmark stage. Type `.py`, size 5.0 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 206 lines; direct main block: yes; imports: `numpy`, `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:load_model_data`, `src.models.utils.evaluation:evaluate_predictions`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>load_predictions</code>	<code>none</code>	Load and combine fixed linear and tree predictions.	See docstring; no additional side effect verified.	21
<code>target_summary</code>	<code>samples</code>	Summarize the target in train and validation samples.	See docstring; no additional side effect verified.	37
<code>prediction_summary</code>	<code>predictions</code>	Summarize prediction distributions for every model and sample.	See docstring; no additional side effect verified.	61
<code>validation_by_year</code>	<code>predictions, train_mean</code>	Evaluate each fixed model separately for every validation year.	See docstring; no additional side effect verified.	87
<code>extreme_predictions</code>	<code>predictions</code>	Return the largest absolute fixed-model predictions.	See docstring; no additional side effect verified.	122
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	140

Critical decisions. Fits fixed specifications on train and reports train/validation metrics; test rows are not used.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.69 File 69: `init .py`

Identity. `src/models/step_01_fixed/__init__.py`. **Role.** Fixed train/validation benchmark stage. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. Fits fixed specifications on train and reports train/validation metrics; test rows are not used.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.70 File 70: `01 tune linear models.py`

Identity. `src/models/step_02_tuning/01_tune_linear_models.py`. **Role.** Annual expanding-window tuning stage. Type `.py`, size 2.5 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 108 lines; direct main block: yes; imports: `numpy`, `pathlib:Path`, `sklearn.cross_decomposition:PLSRegression`, `sklearn.linear_model:ElasticNet`, `sklearn.pipeline:make_pipeline`, `sklearn.preprocessing:StandardScaler`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:expanding_year_folds`, `load_model_data`, `src.models.utils.estimation:save_results`, `tune_grid`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>make_model</code>	<code>family, params</code>	Construct one linear-model candidate.	See docstring; no additional side effect verified.	40
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	64

Critical decisions. Loads train only, uses annual expanding folds beginning in 2005, searches an explicit grid, and saves the minimum-MSE parameters.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.71 File 71: 02 tune tree models.py

Identity. `src/models/step_02_tuning/02_tune_tree_models.py`. **Role.** Annual expanding-window tuning stage. Type `.py`, size 2.6 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 114 lines; direct main block: yes; imports: `pathlib:Path`, `sklearn.ensemble:RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR`, `TARGET`, `src.models.utils.data:expanding_year_folds`, `load_model_data`, `src.models.utils.estimation:save_results`, `tune_grid`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>make_model</code>	<code>family, params</code>	Construct one tree-model candidate.	See docstring; no additional side effect verified.	55
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	70

Critical decisions. Loads train only, uses annual expanding folds beginning in 2005, searches an explicit grid, and saves the minimum-MSE parameters.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.72 File 72: 03 compare tuning results.py

Identity. `src/models/step_02_tuning/03_compare_tuning_results.py`. **Role.** Annual expanding-window tuning stage. Type `.py`, size 1.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 50 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	22

Critical decisions. Loads train only, uses annual expanding folds beginning in 2005, searches an explicit grid, and saves the minimum-MSE parameters.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.73 File 73: `init .py`

Identity. `src/models/step_02_tuning/__init__.py`. **Role.** Annual expanding-window tuning stage. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. Loads train only, uses annual expanding folds beginning in 2005, searches an explicit grid, and saves the minimum-MSE parameters.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.74 File 74: `01 optimized linear models.py`

Identity. `src/models/step_03_optimization/01_optimized_linear_models.py`. **Role.** Optimized train/validation comparison stage. Type `.py`, size 4.5 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 176 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_model, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	<code>all_predictors</code>	Define the optimized linear model specifications.	See docstring; no additional side effect verified.	30
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	80

Critical decisions. Loads tuned parameters, fits train, compares on validation, and rejects test rows in comparison files.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.75 File 75: 02 optimized tree models.py

Identity. `src/models/step_03_optimization/02_optimized_tree_models.py`. **Role.** Optimized train/validation comparison stage. Type `.py`, size 3.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 97 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `sklearn.ensemble:RandomForestRegressor`, `sklearn.tree:DecisionTreeRegressor`, `src.config:MODEL_OUTPUT_DIR,TARGET`, `src.models.utils.data:arrays`, `load_model_data`, `src.models.utils.estimation:load_best_parameters`, `src.models.utils.evaluation:evaluate_model`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>optimized_models</code>	none	Define tree models from the saved tuning results.	See docstring; no additional side effect verified.	23
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	49

Critical decisions. Loads tuned parameters, fits train, compares on validation, and rejects test rows in comparison files.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.76 File 76: 03 compare optimized models.py

Identity. `src/models/step_03_optimization/03_compare_optimized_models.py`. **Role.** Optimized train/validation comparison stage. Type `.py`, size 1.9 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 82 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	26

Critical decisions. Loads tuned parameters, fits train, compares on validation, and rejects test rows in comparison files.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.77 File 77: 04 compare fixed vs optimized.py

Identity. `src/models/step_03_optimization/04_compare_fixed_vs_optimized.py`. **Role.** Optimized train/validation comparison stage. Type `.py`, size 6.0 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 231 lines; direct main block: yes; imports: `pandas`, `pathlib:Path`, `src.config:MODEL_OUTPUT_DIR`, `src.models.utils.evaluation:rank_models`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	31

Critical decisions. Loads tuned parameters, fits train, compares on validation, and rejects test rows in comparison files.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.78 File 78: `init .py`

Identity. `src/models/step_03_optimization/__init__.py`. **Role.** Optimized train/validation comparison stage. Type `.py`, size 0.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 0 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. Loads tuned parameters, fits train, compares on validation, and rejects test rows in comparison files.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.79 File 79: `01 time series robustness.py`

Identity. `src/models/step_04_robustness/01_time_series_robustness.py`. **Role.** Time-series robustness stage. Type `.py`, size 4.6 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 164 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.ensemble:RandomForestRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:load_model_data, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_predictions, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_factories</code>	<code>none</code>	Return a fresh optimized Random Forest for each annual refit.	See docstring; no additional side effect verified.	24
<code>walk_forward</code>	<code>data, predictors, model_name, make_model, window</code>	Generate annual validation forecasts using past observations only.	See docstring; no additional side effect verified.	49
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	87

Critical decisions. Compares expanding with 120-month rolling Random Forest refits over validation years 2015-2019.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.80 File 80: `init .py`

Identity. `src/models/step_04_robustness/__init__.py`. **Role.** Time-series robustness stage. Type `.py`, size 36.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 1 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. Compares expanding with 120-month rolling Random Forest refits over validation years 2015-2019.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.81 File 81: `01 final test evaluation.py`

Identity. `src/models/step_05_test/01_final_test_evaluation.py`. **Role.** One-time final test estimation and output stage. Type `.py`, size 13.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 586 lines; direct main block: yes; imports: `numpy, pandas, pathlib:Path, sklearn.cross_decomposition:PLSRegression, sklearn.ensemble:RandomForestRegressor, sklearn.linear_model:ElasticNet, LinearRegression, sklearn.pipeline:make_pipeline, sklearn.preprocessing:StandardScaler, sklearn.tree:DecisionTreeRegressor, src.config:MODEL_OUTPUT_DIR, TARGET, src.models.utils.data:arrays, load_model_data, ols3_predictors, src.models.utils.estimation:load_best_parameters, src.models.utils.evaluation:evaluate_model, rank_models, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>model_specifications</code>	<code>predictors</code>	Return the final models selected using validation results.	See docstring; no additional side effect verified.	40
<code>linear_coefficient_table</code>	<code>model_name</code> , <code>model</code> , <code>predictors</code>	Extract coefficients from a fitted linear-model pipeline.	See docstring; no additional side effect verified.	122
<code>tree_feature_importance_table</code>	<code>model_name</code> , <code>model</code> , <code>predictors</code>	Extract impurity-based importance from a fitted tree model.	See docstring; no additional side effect verified.	169
<code>pls_component_table</code>	<code>model</code>	Summarize fitted PLS component-score variation.	See docstring; no additional side effect verified.	205
<code>model_complexity_row</code>	<code>model_name</code> , <code>model</code> , <code>predictors</code>	Create one model-complexity summary row.	See docstring; no additional side effect verified.	242
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	324

Critical decisions. Refits frozen specifications on development data and saves only test metrics/predictions plus interpretation inputs.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.82 File 82: `init .py`

Identity. `src/models/step_05_test/__init__.py`. **Role.** One-time final test estimation and output stage. Type `.py`, size 24.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 1 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.83 File 83: `00 utils.py`

Identity. `src/models/step_06_interpretation/00_utils.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 12.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 588 lines; direct main block: no; imports: `numpy, pandas, src.config:INTERPRETATION_OUTPUT_DIR, MODEL_OUTPUT_DIR, src.models.utils.evaluation:monthly_mse`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>require_file</code>	<code>path, prior_step</code>	Return a required path or explain which prior step must run.	See docstring; no additional side effect verified.	40
<code>require_columns</code>	<code>data, columns, path</code>	Check that one input contains all required columns.	See docstring; no additional side effect verified.	51
<code>read_csv</code>	<code>path, prior_step, columns</code>	Read one required CSV file and validate its columns.	See docstring; no additional side effect verified.	61
<code>read_parquet</code>	<code>path, prior_step, columns</code>	Read one required Parquet file and validate its columns.	See docstring; no additional side effect verified.	76
<code>ensure_unique</code>	<code>data, columns, label</code>	Reject duplicated report keys.	See docstring; no additional side effect verified.	91
<code>save_csv</code>	<code>data, filename</code>	Write one Step 6 CSV output.	See docstring; no additional side effect verified.	99
<code>load_test_outputs</code>	<code>none</code>	Load final test metrics, ranking, and predictions.	See docstring; no additional side effect verified.	112
<code>load_interpretation_inputs</code>	<code>none</code>	Load model information saved during Step 5 fitting.	See docstring; no additional side effect verified.	244
<code>load_robustness_outputs</code>	<code>none</code>	Load validation-only robustness results.	See docstring; no additional side effect verified.	411
<code>yearly_prediction_metrics</code>	<code>predictions</code>	Calculate test performance separately by model and year.	See docstring; no additional side effect verified.	458
<code>add_display_names</code>	<code>data</code>	Add readable model names to an output table.	See docstring; no additional side effect verified.	578

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.84 File 84: 01 model interpretation.py

Identity. `src/models/step_06_interpretation/01_model_interpretation.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 7.8 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 364 lines; direct main block: yes; imports: `importlib:import_module`, `pandas`, `pathlib:Path`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	none	Create model-level and yearly final-test summaries.	Writes/fits only if stated by the function or enclosing main workflow.	22

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.85 File 85: 02 feature importance.py

Identity. `src/models/step_06_interpretation/02_feature_importance.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 6.4 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 320 lines; direct main block: yes; imports: `importlib:import_module`, `pandas`, `pathlib:Path`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>top_coefficient_table</code>	<code>coefficients, model_name</code>	Create the unified top-coefficient format for one model.	See docstring; no additional side effect verified.	21
<code>top_tree_importance_table</code>	<code>tree_importance, model_name</code>	Create the unified top tree-importance format.	See docstring; no additional side effect verified.	68
<code>main</code>	<code>none</code>	Load Step 5 interpretation inputs and create report tables.	Writes/fits only if stated by the function or enclosing main workflow.	102

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.86 File 86: 03 create result tables.py

Identity. `src/models/step_06_interpretation/03_create_result_tables.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 13.3 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 636 lines; direct main block: yes; imports: `ast:literal_eval, importlib:import_module, json, openpyxl.styles:Alignment,Font,PatternFill, openpyxl.utils.dataframe:dataframe_to_rows, openpyxl.utils:get_column_letter, openpyxl:Workbook, pandas, pathlib:Path, sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>final_results</code>	<code>none</code>	Load the final-test ranking created in Step 6 file 01.	See docstring; no additional side effect verified.	32
<code>fixed_vs_optimized</code>	<code>none</code>	Compare fixed and optimized validation results.	See docstring; no additional side effect verified.	76
<code>best_hyperparameters</code>	<code>none</code>	Load the selected tuning parameters without rebuilding models.	See docstring; no additional side effect verified.	215
<code>yearly_results</code>	<code>none</code>	Load yearly final-test results created in Step 6 file 01.	See docstring; no additional side effect verified.	272
<code>complexity_summary</code>	<code>none</code>	Load fitted complexity measures saved from the Step 5 models.	See docstring; no additional side effect verified.	314
<code>top_features</code>	<code>none</code>	Load the unified top-feature table from Step 6 file 02.	See docstring; no additional side effect verified.	374
<code>behavior_flags</code>	<code>none</code>	Load model-behavior indicators created in Step 6 file 01.	See docstring; no additional side effect verified.	415
<code>format_worksheet</code>	<code>sheet, data</code>	Apply simple academic formatting to one worksheet.	See docstring; no additional side effect verified.	445
<code>write_workbook</code>	<code>tables, path</code>	Write all report tables to one formatted workbook.	See docstring; no additional side effect verified.	541
<code>main</code>	<code>none</code>	Create all report-ready tables.	Writes/fits only if stated by the function or enclosing main workflow.	569

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.87 File 87: 04 create figures.py

Identity. `src/models/step_06_interpretation/04_create_figures.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 12.0 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 677 lines; direct main block: yes; imports: `importlib:import_module`, `matplotlib`, `matplotlib.pyplot`, `numpy`, `os`, `pathlib:Path`, `re`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>save_figure</code>	<code>filename</code>	Save and close the current Matplotlib figure.	See docstring; no additional side effect verified.	36
<code>display_names</code>	<code>models</code>	Return readable names for a sequence of model identifiers.	See docstring; no additional side effect verified.	54
<code>safe_filename</code>	<code>value</code>	Convert a model name into a safe filename component.	See docstring; no additional side effect verified.	65
<code>parse_boolean</code>	<code>series</code>	Convert CSV boolean values reliably to True and False.	See docstring; no additional side effect verified.	74
<code>bar_figure</code>	<code>data, value, filename, title, ascending, zero_line</code>	Create one sorted model-comparison bar chart.	See docstring; no additional side effect verified.	95
<code>yearly_lines</code>	<code>yearly, value, filename, title, zero_line</code>	Create one yearly model-performance line chart.	See docstring; no additional side effect verified.	144
<code>prediction_scatter</code>	<code>results, predictions</code>	Create realized-versus-predicted scatter for the best model.	See docstring; no additional side effect verified.	193
<code>prediction_distributions</code>	<code>predictions</code>	Compare the prediction distributions of fitted models.	See docstring; no additional side effect verified.	274
<code>random_forest_importance_figure</code>	<code>none</code>	Plot the most important Random Forest predictors.	See docstring; no additional side effect verified.	332
<code>decision_tree_importance_figure</code>	<code>none</code>	Plot the most important Decision Tree predictors.	See docstring; no additional side effect verified.	382
<code>pls_coefficient_figure</code>	<code>none</code>	Plot the largest PLS coefficients by absolute magnitude.	See docstring; no additional side effect verified.	432
<code>elastic_net_coefficient_figure</code>	<code>none</code>	Plot nonzero Elastic Net coefficients or save an explanatory note.	See docstring; no additional side effect verified.	482
<code>main</code>	<code>none</code>	Create all report figures from saved Step 6 outputs.	Writes/fits only if stated by the function or enclosing main workflow.	575

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.88 File 88: 05 report notes.py

Identity. `src/models/step_06_interpretation/05_report_notes.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 14.1 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 506 lines; direct main block: yes; imports: `importlib:import_module`, `pandas`, `pathlib:Path`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>metric</code>	<code>value</code>	Format a numerical report value.	See docstring; no additional side effect verified.	18
<code>integer_metric</code>	<code>value</code>	Format a count without decimal places.	See docstring; no additional side effect verified.	26
<code>parse_boolean</code>	<code>series</code>	Convert CSV boolean values reliably.	See docstring; no additional side effect verified.	34
<code>display_name</code>	<code>model</code>	Return the readable name of one model.	See docstring; no additional side effect verified.	55
<code>main</code>	<code>none</code>	Create automatically generated interpretation notes.	Writes/fits only if stated by the function or enclosing main workflow.	63

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.89 File 89: README.md

Identity. `src/models/step_06_interpretation/README.md`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.md`, size 2.1 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.90 File 90: `init .py`

Identity. `src/models/step_06_interpretation/__init__.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 61.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 1 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.91 File 91: `run all.py`

Identity. `src/models/step_06_interpretation/run_all.py`. **Role.** Frozen-output interpretation and report-artifact stage. Type `.py`, size 569.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 25 lines; direct main block: yes; imports: `pathlib:Path`, `subprocess`, `sys`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>main</code>	<code>none</code>	No function docstring.	Writes/fits only if stated by the function or enclosing main workflow.	16

Critical decisions. Reads frozen outputs, validates schemas/model membership, and produces tables, figures, workbook, or narrative notes without refitting models.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.92 File 92: `init .py`

Identity. `src/models/utils/__init__.py`. **Role.** Shared model loading, estimation, or evaluation utility. Type `.py`, size 50.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 1 lines; direct main block: no; imports: none.

No functions are defined.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.93 File 93: `data.py`

Identity. `src/models/utils/data.py`. **Role.** Shared model loading, estimation, or evaluation utility. Type `.py`, size 1.9 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 84 lines; direct main block: no; imports: `numpy, pandas, src.config:TARGET, VALIDATION_END, CLEAN_FULL_FILE, CLEAN_PREDICTOR_FILE, TRAIN_END`.

Function	Inputs	Return/output meaning	Side effects	Line
load_model_data	sample_names	Load the model data and return the requested samples.	See docstring; no additional side effect verified.	17
arrays	samples, predictors, target	Convert each sample into predictor and target arrays.	See docstring; no additional side effect verified.	46
expanding_year_folds	data, first_validation_year	Create annual expanding-window validation folds.	See docstring; no additional side effect verified.	57
ols3_predictors	none	Return size, book-to-market, and momentum predictors.	See docstring; no additional side effect verified.	75

Critical decisions. Defines train through 2014, validation 2015-2019, development through 2019, test 2020-2025, and annual expanding folds.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.94 File 94: `estimation.py`

Identity. `src/models/utils/estimation.py`. **Role.** Shared model loading, estimation, or evaluation utility. Type `.py`, size 4.4 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 136 lines; direct main block: no; imports: `ast:literal_eval`, `numpy`, `pandas`, `re`, `sklearn.model_selection:ParameterGrid`, `src.config:TARGET`, `src.models.utils.data:arrays`, `src.models.utils.evaluation:evaluate_model`, `monthly_mse`.

Function	Inputs	Return/output meaning	Side effects	Line
<code>fit_models</code>	<code>models, samples, predictors, target, effect</code>	Fit models on train and evaluate train and validation.	See docstring; no additional side effect verified.	15
<code>tune_grid</code>	<code>family, grid, make_model, data, predictors, folds, target</code>	Test all hyperparameter combinations and select the lowest CV MSE.	See docstring; no additional side effect verified.	60
<code>save_results</code>	<code>family, results, best_params, results_file, parameters_file</code>	Save tuning results and the best parameters.	See docstring; no additional side effect verified.	103
<code>load_best_parameters</code>	<code>path, required_families</code>	Load one parameter dictionary or a dictionary by family.	See docstring; no additional side effect verified.	118
<code>parse</code>	<code>value</code>	No function docstring.	See docstring; no additional side effect verified.	122

Critical decisions. Grid search chooses the minimum mean monthly MSE; no one-standard-error rule is implemented.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.95 File 95: `evaluation.py`

Identity. `src/models/utils/evaluation.py`. **Role.** Shared model loading, estimation, or evaluation utility. Type `.py`, size 2.9 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

Execution. 106 lines; direct main block: no; imports: `numpy`, `pandas`, `sklearn.metrics:mean_absolute_error,mean_squared_error`.

Function	Inputs	Return/output meaning	Side effects	Line
monthly_mse	y_true, y_pred, months	Calculate MSE for each month and average equally across months.	See docstring; no additional side effect verified.	8
evaluate_predictions	y_true, y_pred, benchmark, months	Calculate prediction metrics.	See docstring; no additional side effect verified.	18
evaluate_model	model_name, samples, predictions, target, benchmark_mean	Evaluate one model and create its prediction table.	See docstring; no additional side effect verified.	51
rank_models	metrics, sample	Rank models by monthly MSE.	See docstring; no additional side effect verified.	96

Critical decisions. Primary loss equally weights months; OOS R-squared uses a constant historical-mean benchmark.

Quality assessment. Chronological separation and pipelines are strong. Main risks are documentation drift, dimensionality, no formal loss-comparison tests, and unpinned dependencies.

C.96 File 96: Makefile

Identity. `thesis/Makefile`. **Role.** Current course-paper source, build artifact, or bibliography. Type `Makefile`, size 226.0 B. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.97 File 97: README.md

Identity. `thesis/README.md`. **Role.** Current course-paper source, build artifact, or bibliography. Type `.md`, size 1.7 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.98 File 98: paper.tex

Identity. `thesis/paper.tex`. **Role.** Current course-paper source, build artifact, or bibliography. Type `.tex`, size 30.2 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

C.99 File 99: references.bib

Identity. `thesis/references.bib`. **Role.** Current course-paper source, build artifact, or bibliography. Type `.bib`, size 3.6 KB. Producer/input relationship: Not verifiable from the current repository.. Consumer/dependent relationship: Not verifiable from the current repository..

No Python function inventory applies. Its execution or compilation behavior follows the file type.

Critical decisions. No separate consequential live decision identified beyond the file's documented role.

Quality assessment. No material leakage risk by itself; review status and staleness according to category.

D Complete Function Inventory

The AST scan found 182 function definitions across active, legacy, and backup Python files; 131 belong to active non-legacy source. Nested helpers are included.

File	Function	Arguments	Line	Purpose
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/01_fixed_linear_model_s.py	fixed_models	all_predictors	28	Define the fixed linear model specifications.
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/01_fixed_linear_model_s.py	main	none	63	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/02_fixed_tree_models.py	optimized_models	none	24	Define the optimized tree-model specifications.
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/02_fixed_tree_models.py	main	none	82	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_01_fixed/03_compare_fixed_models.py	main	none	18	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/01_optimized_linear_models.py	optimized_models	all_predictors	30	Define the optimized linear model specifications.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/01_optimized_linear_models.py	main	none	80	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/02_optimized_tree_models.py	optimized_models	parameters	21	Construct tree models from validation-selected parameters.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/02_optimized_tree_models.py	main	none	52	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/03_compare_optimized_models.py	main	none	18	No function docstring.
backup/model_reorganization_20260719_054538/src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	main	none	19	No function docstring.
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	fit_models	models, samples, predictors, target	16	Fit models on development data and evaluate their predictions.

File	Function	Arguments	Line	Purpose
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	tune_grid	family, grid, make_model, data, predictors, folds, target	53	Test all hyperparameter combinations and select the lowest CV MSE.
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	save_results	family, results, best_params, results_file, parameters_file	120	Save tuning results and the best parameters.
backup/model_reorganization_20260719_054538/src/models/utils/estimation.py	load_best_parameters	path	138	Load the saved best parameter dictionary.
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	monthly_mse	y_true, y_pred, months	8	Calculate MSE for each month and average equally across months.
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	evaluate_predictions	y_true, y_pred, benchmark, months	18	Calculate prediction metrics.
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	evaluate_model	model_name, samples, predictions, target, benchmark_mean	51	Evaluate one model and create its prediction table.
backup/model_reorganization_20260719_054538/src/models/utils/evaluation.py	rank_models	metrics, sample	96	Rank models by monthly MSE.
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/02_fixed_tree_models.py	fixed_models	none	21	Define the fixed tree-model specifications.
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/02_fixed_tree_models.py	main	none	46	No function docstring.
backup/remove_gradient_boosting_20260719_063836/src/models/step_01_fixed/03_compare_fixed_models.py	main	none	18	No function docstring.
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/02_tune_tree_models.py	make_model	family, params	74	Construct one tree-model candidate.
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/02_tune_tree_models.py	main	none	96	No function docstring.
backup/remove_gradient_boosting_20260719_063836/src/models/step_02_tuning/03_compare_tuning_results.py	main	none	23	No function docstring.
backup/remove_gradient_boosting_20260719_063836/src/models/step_03_optimization/02_optimized_tree_models.py	optimized_models	none	23	Define tree models from the saved tuning results.

File	Function	Arguments	Line	Purpose
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_03_optimization/02_optimiz ed_tree_models.py	main	none	60	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_03_optimization/03_compare _optimized_models.py	main	none	18	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_03_optimization/04_compare _fixed_vs_optimized.py	main	none	19	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_04_robustness/01_time_seri es_robustness.py	model_factories	none	25	Return fresh optimized tree models for each annual refit.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_04_robustness/01_time_seri es_robustness.py	walk_forward	data, predictors, model_name, make_model, window	56	Generate annual validation forecasts using past observations only.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_04_robustness/01_time_seri es_robustness.py	main	none	86	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_05_test/01_final_test_eval uation.py	model_specifications	predictors	28	Return the prespecified fixed and optimized models.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_05_test/01_final_test_eval uation.py	main	none	144	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/01_portfolio_ sorts.py	prediction_groups	none	21	Yield final-test model predictions.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/01_portfolio_ sorts.py	main	none	48	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/02_compare_po rtfolio_results.py	select_columns	data	26	No function docstring.
backup/remove_gradient_boostin g_20260719_063836/src/models/s tep_06_portfolio/02_compare_po rtfolio_results.py	main	none	30	No function docstring.

File	Function	Arguments	Line	Purpose
backup/remove_gradient_boosting_20260719_063836/src/models/step_06_portfolio/03_final_model_and_portfolio_summary.py	main	none	18	No function docstring.
src/acquisition/01_create_locked_stock_universe.py	clean_ticker	value	28	Convert ticker to Yahoo format, for example BRK.B -> BRK-B.
src/acquisition/01_create_locked_stock_universe.py	normalize_columns	table	41	Return a copy with simple string column names.
src/acquisition/01_create_locked_stock_universe.py	find_column	table, words	55	Find the first column containing all requested words.
src/acquisition/01_create_locked_stock_universe.py	build_sp500_universe	none	64	Combine current and historical Wikipedia S&P 500 tickers.
src/acquisition/01_create_locked_stock_universe.py	main	none	108	No function docstring.
src/acquisition/02_create_fama_french_rf_input.py	download_fama_french_factors	none	19	Download and parse monthly Fama-French three-factor data.
src/acquisition/02_create_fama_french_rf_input.py	main	none	52	No function docstring.
src/acquisition/03_download_market_inputs.py	download_yahoo_daily	ticker	26	Download one Yahoo daily price series with standard columns.
src/acquisition/03_download_market_inputs.py	save_market_input	ticker, output_file, source_name	56	Download, save, and register one market input.
src/acquisition/03_download_market_inputs.py	main	none	75	No function docstring.
src/acquisition/04_create_welch_goyal_input.py	load_and_validate_welch_goyal	none	22	Validate the local cleaned Welch-Goyal monthly input.
src/acquisition/04_create_welch_goyal_input.py	main	none	49	No function docstring.
src/acquisition/manifest.py	update_input_manifest	manifest_file, input_file, source, coverage_start, coverage_end, notes	18	Insert or replace one manifest row for a permanent input file.
src/data/01_build_clean_yahoo_daily.py	load_locked_universe	none	32	Load unique Yahoo-format tickers.
src/data/01_build_clean_yahoo_daily.py	download_daily_prices	tickers	53	Download prices and return the data and failed ticker list.
src/data/01_build_clean_yahoo_daily.py	clean_daily_prices	daily	126	Remove bad rows and ticker histories with excessive bad data.
src/data/01_build_clean_yahoo_daily.py	main	none	217	Download, clean, and save the daily stock dataset.
src/data/02_build_monthly_stock_features.py	build_monthly_panel	daily	30	Aggregate daily prices into monthly stock characteristics.
src/data/02_build_monthly_stock_features.py	remove_bad_monthly_tickers	monthly	85	Remove tickers with repeated bad returns and drop isolated bad months.
src/data/02_build_monthly_stock_features.py	compound	returns	147	Compound simple returns.
src/data/02_build_monthly_stock_features.py	add_stock_features	monthly	152	Create momentum, volatility, liquidity, and trend predictors.

File	Function	Arguments	Line	Purpose
src/data/02_build_monthly_stoc k_features.py	load_market_data	none	209	Load permanent daily S&P 500 and VIX files.
src/data/02_build_monthly_stoc k_features.py	add_market_features	monthly, market	235	Add monthly market return, market volatility, and VIX variables.
src/data/02_build_monthly_stoc k_features.py	add_beta_idiovol	monthly, daily, market	286	Create rolling market beta and idiosyncratic volatility.
src/data/02_build_monthly_stoc k_features.py	load_fama_french_rf	none	347	Load the permanent monthly risk-free rate.
src/data/02_build_monthly_stoc k_features.py	add_target	monthly, rf	365	Create next-month stock and excess returns.
src/data/02_build_monthly_stoc k_features.py	main	none	393	Build and save the monthly stock panel.
src/data/03_add_fundamentals_a nd_macro.py	column	data, name	25	Return a column or a missing-value series.
src/data/03_add_fundamentals_a nd_macro.py	ratio	numerator, denominator	33	Calculate a ratio safely.
src/data/03_add_fundamentals_a nd_macro.py	clean_compustat	none	50	Create compact annual accounting predictors with a six-month lag.
src/data/03_add_fundamentals_a nd_macro.py	merge_compustat	monthly, comp	374	Attach the latest available annual Compustat report.
src/data/03_add_fundamentals_a nd_macro.py	main	none	408	Merge annual Compustat fundamentals into the monthly panel.
src/data/04_build_raw_kelly_da taset.py	add_industry_dummies	data	37	Create SIC2 industry dummies.
src/data/04_build_raw_kelly_da taset.py	add_welch_goyal_macro	data	62	Merge monthly Welch-Goyal macro variables.
src/data/04_build_raw_kelly_da taset.py	validate_columns	data, columns, label	87	Confirm that the required variables are available.
src/data/04_build_raw_kelly_da taset.py	prepare_raw_data	data, characteristics, market_variables, macro_variables, sic2_dummies	103	Select and save the raw base predictor panel.
src/data/04_build_raw_kelly_da taset.py	main	none	173	Build and save the raw Kelly-style base dataset.
src/data/05_clean_and_rank_nor malize.py	target_diagnostics	data	37	Document unusually large target returns.
src/data/05_clean_and_rank_nor malize.py	impute_predictors	data, predictors	71	Impute missing predictors using monthly cross-sectional medians.
src/data/05_clean_and_rank_nor malize.py	rank_characteristics	data, characteristics	108	Map monthly characteristic ranks to the interval [-1, 1].
src/data/05_clean_and_rank_nor malize.py	create_interactions	data, characteristics, macro_variables	134	Create ranked-characteristic and macro interactions.
src/data/05_clean_and_rank_nor malize.py	check_final_data	data, predictors, characteristics	169	Check the final model-ready dataset.
src/data/05_clean_and_rank_nor malize.py	main	none	210	Create and save the final model-ready dataset.
src/models/legacy/gradient_boo sting/source_snapshots/step_01 _fixed/02_fixed_tree_models.py	fixed_models	none	21	Define the fixed tree-model specifications.

File	Function	Arguments	Line	Purpose
src/models/legacy/gradient_boosting/source_snapshots/step_01_fixed/02_fixed_tree_models.py	main	none	46	No function docstring.
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/02_tune_tree_models.py	make_model	family, params	74	Construct one tree-model candidate.
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/02_tune_tree_models.py	main	none	96	No function docstring.
src/models/legacy/gradient_boosting/source_snapshots/step_02_tuning/03_compare_tuning_results.py	main	none	23	No function docstring.
src/models/legacy/gradient_boosting/source_snapshots/step_03_optimization/02_optimized_tree_models.py	optimized_models	none	23	Define tree models from the saved tuning results.
src/models/legacy/gradient_boosting/source_snapshots/step_03_optimization/02_optimized_tree_models.py	main	none	60	No function docstring.
src/models/legacy/gradient_boosting/source_snapshots/step_04_robustness/01_time_series_robustness.py	model_factories	none	25	Return fresh optimized tree models for each annual refit.
src/models/legacy/gradient_boosting/source_snapshots/step_04_robustness/01_time_series_robustness.py	walk_forward	data, predictors, model_name, make_model, window	56	Generate annual validation forecasts using past observations only.
src/models/legacy/gradient_boosting/source_snapshots/step_04_robustness/01_time_series_robustness.py	main	none	86	No function docstring.
src/models/legacy/gradient_boosting/source_snapshots/step_05_test/01_final_test_evaluation.py	model_specifications	predictors	28	Return the prespecified fixed and optimized models.
src/models/legacy/gradient_boosting/source_snapshots/step_05_test/01_final_test_evaluation.py	main	none	144	No function docstring.
src/models/step_01_fixed/01_fixed_linear_models.py	fixed_models	all_predictors	28	Define the fixed linear model specifications.
src/models/step_01_fixed/01_fixed_linear_models.py	main	none	63	No function docstring.
src/models/step_01_fixed/02_fixed_tree_models.py	fixed_models	none	21	Define the fixed tree-model specifications.
src/models/step_01_fixed/02_fixed_tree_models.py	main	none	41	No function docstring.

File	Function	Arguments	Line	Purpose
src/models/step_01_fixed/03_compare_fixed_models.py	main	none	26	No function docstring.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	load_predictions	none	21	Load and combine fixed linear and tree predictions.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	target_summary	samples	37	Summarize the target in train and validation samples.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	prediction_summary	predictions	61	Summarize prediction distributions for every model and sample.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	validation_by_year	predictions, train_mean	87	Evaluate each fixed model separately for every validation year.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	extreme_predictions	predictions	122	Return the largest absolute fixed-model predictions.
src/models/step_01_fixed/04_diagnostics_fixed_models.py	main	none	140	No function docstring.
src/models/step_02_tuning/01_tune_linear_models.py	make_model	family, params	40	Construct one linear-model candidate.
src/models/step_02_tuning/01_tune_linear_models.py	main	none	64	No function docstring.
src/models/step_02_tuning/02_tune_tree_models.py	make_model	family, params	55	Construct one tree-model candidate.
src/models/step_02_tuning/02_tune_tree_models.py	main	none	70	No function docstring.
src/models/step_02_tuning/03_compare_tuning_results.py	main	none	22	No function docstring.
src/models/step_03_optimization/01_optimized_linear_models.py	optimized_models	all_predictors	30	Define the optimized linear model specifications.
src/models/step_03_optimization/01_optimized_linear_models.py	main	none	80	No function docstring.
src/models/step_03_optimization/02_optimized_tree_models.py	optimized_models	none	23	Define tree models from the saved tuning results.
src/models/step_03_optimization/02_optimized_tree_models.py	main	none	49	No function docstring.
src/models/step_03_optimization/03_compare_optimized_models.py	main	none	26	No function docstring.
src/models/step_03_optimization/04_compare_fixed_vs_optimized.py	main	none	31	No function docstring.
src/models/step_04_robustness/01_time_series_robustness.py	model_factories	none	24	Return a fresh optimized Random Forest for each annual refit.
src/models/step_04_robustness/01_time_series_robustness.py	walk_forward	data, predictors, model_name, make_model, window	49	Generate annual validation forecasts using past observations only.
src/models/step_04_robustness/01_time_series_robustness.py	main	none	87	No function docstring.
src/models/step_05_test/01_final_test_evaluation.py	model_specifications	predictors	40	Return the final models selected using validation results.
src/models/step_05_test/01_final_test_evaluation.py	linear_coefficient_table	model_name, model, predictors	122	Extract coefficients from a fitted linear-model pipeline.

File	Function	Arguments	Line	Purpose
src/models/step_05_test/01_final_test_evaluation.py	tree_feature_importance_table	model_name, model, predictors	169	Extract impurity-based importance from a fitted tree model.
src/models/step_05_test/01_final_test_evaluation.py	pls_component_table	model	205	Summarize fitted PLS component-score variation.
src/models/step_05_test/01_final_test_evaluation.py	model_complexity_row	model_name, model, predictors	242	Create one model-complexity summary row.
src/models/step_05_test/01_final_test_evaluation.py	main	none	324	No function docstring.
src/models/step_06_interpretation/00_utils.py	require_file	path, prior_step	40	Return a required path or explain which prior step must run.
src/models/step_06_interpretation/00_utils.py	require_columns	data, columns, path	51	Check that one input contains all required columns.
src/models/step_06_interpretation/00_utils.py	read_csv	path, prior_step, columns	61	Read one required CSV file and validate its columns.
src/models/step_06_interpretation/00_utils.py	read_parquet	path, prior_step, columns	76	Read one required Parquet file and validate its columns.
src/models/step_06_interpretation/00_utils.py	ensure_unique	data, columns, label	91	Reject duplicated report keys.
src/models/step_06_interpretation/00_utils.py	save_csv	data, filename	99	Write one Step 6 CSV output.
src/models/step_06_interpretation/00_utils.py	load_test_outputs	none	112	Load final test metrics, ranking, and predictions.
src/models/step_06_interpretation/00_utils.py	load_interpretation_inputs	none	244	Load model information saved during Step 5 fitting.
src/models/step_06_interpretation/00_utils.py	load_robustness_outputs	none	411	Load validation-only robustness results.
src/models/step_06_interpretation/00_utils.py	yearly_prediction_metrics	predictions	458	Calculate test performance separately by model and year.
src/models/step_06_interpretation/00_utils.py	add_display_names	data	578	Add readable model names to an output table.
src/models/step_06_interpretation/01_model_interpretation.py	main	none	22	Create model-level and yearly final-test summaries.
src/models/step_06_interpretation/02_feature_importance.py	top_coefficient_table	coefficients, model_name	21	Create the unified top-coefficient format for one model.
src/models/step_06_interpretation/02_feature_importance.py	top_tree_importance_table	tree_importance, model_name	68	Create the unified top tree-importance format.
src/models/step_06_interpretation/02_feature_importance.py	main	none	102	Load Step 5 interpretation inputs and create report tables.
src/models/step_06_interpretation/03_create_result_tables.py	final_results	none	32	Load the final-test ranking created in Step 6 file 01.
src/models/step_06_interpretation/03_create_result_tables.py	fixed_vs_optimized	none	76	Compare fixed and optimized validation results.
src/models/step_06_interpretation/03_create_result_tables.py	best_hyperparameters	none	215	Load the selected tuning parameters without rebuilding models.
src/models/step_06_interpretation/03_create_result_tables.py	yearly_results	none	272	Load yearly final-test results created in Step 6 file 01.
src/models/step_06_interpretation/03_create_result_tables.py	complexity_summary	none	314	Load fitted complexity measures saved from the Step 5 models.

File	Function	Arguments	Line	Purpose
src/models/step_06_interpretation/03_create_result_tables.py	top_features	none	374	Load the unified top-feature table from Step 6 file 02.
src/models/step_06_interpretation/03_create_result_tables.py	behavior_flags	none	415	Load model-behavior indicators created in Step 6 file 01.
src/models/step_06_interpretation/03_create_result_tables.py	format_worksheet	sheet, data	445	Apply simple academic formatting to one worksheet.
src/models/step_06_interpretation/03_create_result_tables.py	write_workbook	tables, path	541	Write all report tables to one formatted workbook.
src/models/step_06_interpretation/03_create_result_tables.py	main	none	569	Create all report-ready tables.
src/models/step_06_interpretation/04_create_figures.py	save_figure	filename	36	Save and close the current Matplotlib figure.
src/models/step_06_interpretation/04_create_figures.py	display_names	models	54	Return readable names for a sequence of model identifiers.
src/models/step_06_interpretation/04_create_figures.py	safe_filename	value	65	Convert a model name into a safe filename component.
src/models/step_06_interpretation/04_create_figures.py	parse_boolean	series	74	Convert CSV boolean values reliably to True and False.
src/models/step_06_interpretation/04_create_figures.py	bar_figure	data, value, filename, title, ascending, zero_line	95	Create one sorted model-comparison bar chart.
src/models/step_06_interpretation/04_create_figures.py	yearly_lines	yearly, value, filename, title, zero_line	144	Create one yearly model-performance line chart.
src/models/step_06_interpretation/04_create_figures.py	prediction_scatter	results, predictions	193	Create realized-versus-predicted scatter for the best model.
src/models/step_06_interpretation/04_create_figures.py	prediction_distributions	predictions	274	Compare the prediction distributions of fitted models.
src/models/step_06_interpretation/04_create_figures.py	random_forest_importance_figure	none	332	Plot the most important Random Forest predictors.
src/models/step_06_interpretation/04_create_figures.py	decision_tree_importance_figure	none	382	Plot the most important Decision Tree predictors.
src/models/step_06_interpretation/04_create_figures.py	pls_coefficient_figure	none	432	Plot the largest PLS coefficients by absolute magnitude.
src/models/step_06_interpretation/04_create_figures.py	elastic_net_coefficient_figure	none	482	Plot nonzero Elastic Net coefficients or save an explanatory note.
src/models/step_06_interpretation/04_create_figures.py	main	none	575	Create all report figures from saved Step 6 outputs.
src/models/step_06_interpretation/05_report_notes.py	metric	value	18	Format a numerical report value.
src/models/step_06_interpretation/05_report_notes.py	integer_metric	value	26	Format a count without decimal places.
src/models/step_06_interpretation/05_report_notes.py	parse_boolean	series	34	Convert CSV boolean values reliably.
src/models/step_06_interpretation/05_report_notes.py	display_name	model	55	Return the readable name of one model.
src/models/step_06_interpretation/05_report_notes.py	main	none	63	Create automatically generated interpretation notes.
src/models/step_06_interpretation/run_all.py	main	none	16	No function docstring.

File	Function	Arguments	Line	Purpose
src/models/utils/data.py	load_model_data	sample_names	17	Load the model data and return the requested samples.
src/models/utils/data.py	arrays	samples, predictors, target	46	Convert each sample into predictor and target arrays.
src/models/utils/data.py	expanding_year_folds	data, first_validation_year	57	Create annual expanding-window validation folds.
src/models/utils/data.py	ols3_predictors	none	75	Return size, book-to-market, and momentum predictors.
src/models/utils/estimation.py	fit_models	models, samples, predictors, target, effect	15	Fit models on train and evaluate train and validation.
src/models/utils/estimation.py	tune_grid	family, grid, make_model, data, predictors, folds, target	60	Test all hyperparameter combinations and select the lowest CV MSE.
src/models/utils/estimation.py	save_results	family, results, best_params, results_file, parameters_file	103	Save tuning results and the best parameters.
src/models/utils/estimation.py	load_best_parameters	path, required_families	118	Load one parameter dictionary or a dictionary by family.
src/models/utils/estimation.py	parse	value	122	No function docstring.
src/models/utils/evaluation.py	monthly_mse	y_true, y_pred, months	8	Calculate MSE for each month and average equally across months.
src/models/utils/evaluation.py	evaluate_predictions	y_true, y_pred, benchmark, months	18	Calculate prediction metrics.
src/models/utils/evaluation.py	evaluate_model	model_name, samples, predictions, target, benchmark_mean	51	Evaluate one model and create its prediction table.
src/models/utils/evaluation.py	rank_models	metrics, sample	96	Rank models by monthly MSE.

E Dataset Schemas

Dataset	Compact schema
Monthly stock panel	ticker, month, last adjusted close, 20 stock/technical fields, four market/VIX fields, beta/beta-squared/idio-vol, next stock return, next RF, next excess return
Fundamental panel	all monthly fields plus Compustat date/availability, SIC2, 27 accounting characteristics, R&D missing and match indicator
Raw Kelly panel	ticker, month, target, 47 stock characteristics, four market variables, eight Welch-Goyal variables, 64 SIC2 dummies
Ranked model panel	ticker, month, target, 123 base predictors plus 376 characteristic-macro interactions = 499 predictors
Test predictions	ticker, month, realized_target, prediction, model, sample

The full 502-column schema is recoverable from the parquet metadata and predictor CSV. It is not duplicated verbatim here because the complete predictor inventory already exists at `output/data/final/predictor_columns_kelly_ranked.csv`.

F Critical Decision Register

ID	Decision	Evidence	Benefit	Risk	Disposition
D01	Locked union of current and recorded historical S&P 500 tickers	src/acquisition/01_create_locked_stock_universe.py:64-105	Broadens coverage beyond current constituents	Wikipedia history and ticker identifiers remain incomplete	Verify
D02	Yahoo download uses adjusted close with auto_adjust=False	src/data/01_build_clean_yahoo_daily.py:71-103	Preserves raw OHLC and adjusted close	Vendor revisions and delisting gaps	Retain with disclosure
D03	Daily ticker removal if bad-row share exceeds 10%	src/config.py:69; data file 01:183-194	Drops severely corrupted histories	Threshold is judgmental; failed downloads are not saved	Verify
D04	Daily returns outside [-0.95, 3.0] are invalid	data file 01:146-149	Flags extreme likely data errors	Could remove genuine events or fail to catch split errors	Reconsider
D05	Ticker dropped after two implausible monthly returns	data file 02:85-126	Targets repeated corruption	Whole-history deletion and gap creation	Reconsider
D06	Monthly return is month-end adjusted-close percentage change	data file 02:54-77	Standard simple return	Next-observation shift is unsafe across missing months	Retain formula; verify continuity
D07	Momentum excludes current return via shift(1)	data file 02:154-169	Prevents overlap with current-month return	Requires complete histories	Retain
D08	Six-, 12-, and 36-month compound windows require full windows	data file 02:159-169	Transparent history requirement	Missing periods are observation windows, not guaranteed calendar windows	Verify
D09	Beta and idiosyncratic volatility use 252 daily observations, minimum 126	data file 02:286-335	Uses approximately one year of data	Residual-variance shortcut is not a full rolling regression	Verify
D10	Target uses group shift(-1) then subtracts shifted RF	data file 02:365-385	Creates next-observation excess return	26 retained transitions skip calendar months	Reconsider urgently
D11	Sample ends 2025-12; price download includes 2026-01	src/config.py:54-58	Makes December 2025 target possible	Depends on stable January 2026 input	Retain
D12	Compustat keeps consolidated industrial standard USD records	data file 03:82-91	Improves comparability	May exclude financial-format observations and create coverage patterns	Retain with disclosure
D13	Accounting availability equals fiscal date plus six months at month-end	data file 03:318-323	Conservative look-ahead protection	Ignores actual filing dates	Retain
D14	Backward as-of merge by ticker	data file 03:374-402	Prevents future statements entering past months	Ticker matching can break across symbol changes	Retain and verify matches
D15	SIC2 missing is an explicit dummy	data file 04:37-59	Preserves unmatched observations	Many sparse indicators increase dimensionality	Retain
D16	Welch-Goyal variables merge on same month	data file 04:62-84	Simple alignment	Contradicts intended one-month macro lag and may use unavailable information	Reconsider urgently
D17	Eight macro series are used; inflation column is ignored	src/feature_definitions.py:83-92	Locks intended state variables	Input contains an unused ninth series	Verify
D18	Missing predictors use monthly cross-sectional medians, then zero	data file 05:71-105	No time-series look-ahead; retains rows	Zero fallback for all-missing month is arbitrary	Retain with sensitivity check
D19	Forty-seven characteristics rank to [-1,1] each month	data file 05:108-131	Robust to outliers and comparable scales	Removes magnitude information	Retain
D20	376 ranked-characteristic x macro interactions	data file 05:134-166	Allows state dependence	Dominates 499-dimensional design and raises overfitting risk	Reconsider

ID	Decision	Evidence	Benefit	Risk	Disposition
D21	Raw target is not winsorized	data file 05:377-417	Preserves economic outcomes	Squared loss is dominated by extremes	Sensitivity check
D22	Train ends 2014; validation ends 2019; test begins 2020	src/config.py:60-63; models/utis/data.py:27-35	Chronological separation	Test has only 72 months	Retain
D23	Tuning is executed on train only, annual folds beginning 2005	tuning files and models/utis/data.py:57-71	Prevents 2015-2019 optimization leakage	Contradicts README claim of folds through 2019	Clarify
D24	Candidate selected by minimum average monthly MSE	models/utis/estimation.py:60-100	Directly optimizes stated loss	No uncertainty-aware one-standard-error rule despite documentation	Clarify or implement later
D25	Linear estimators use StandardScaler inside sklearn pipelines	fixed/tuning/test linear scripts	Scaler fits only on model training data	Cross-sectional ranks are scaled again	Retain
D26	Tree models use unscaled 499 predictors	tree scripts	Scaling unnecessary for splits	Sparse/high-dimensional importance is unstable	Retain with caution
D27	Random seed 42 for stochastic estimators	tree, robustness, and test scripts	Reproducible conditional on versions/threading	Dependencies are unpinned	Retain and lock environment
D28	Monthly MSE averages firm MSE equally over months	models/utis/evaluation.py:8-15	Each month has equal weight	Differs slightly from pooled MSE when counts vary	Retain
D29	Historical mean for final test is the development target mean	step_05_test:324-374	Allowable constant benchmark	Not recursively updated during test	Retain and describe
D30	Final models refit on 1990-2019 then predict 2020-2025 once	step_05_test:324-524	Clean final evaluation	Whether researchers avoided viewing test during development is not verifiable	Process control

G Risk Register

ID	Severity	Dimension	Risk	Evidence	Response
I01	High	Validity	26 targets cross gaps and are not next-calendar-month returns	Monthly panel plus data file 02 shift(-1)	Require the next observed month to equal month+1 before assigning target
I02	High	Validity	Same-month macro merge contradicts intended one-month lag	data file 04:62-84	Verify publication timing and lag macro inputs explicitly
I03	High	Reproducibility	run_pipeline.sh calls five missing scripts	Direct existence check	Update wrapper after deciding canonical stages
I04	High	Reproducibility	README tuning period and one-standard-error claim do not match code	models README versus tuning/estimation code	Correct documentation or implementation
I05	Medium	Reproducibility	Failed Yahoo downloads are printed but not persisted	data file 01 main	Save an acquisition failure report
I06	Medium	Validity	Wikipedia union cannot guarantee point-in-time membership or delisting returns	Universe acquisition logic	Use CRSP permanent identifiers for a future version
I07	Medium	Validity	Ticker-based Compustat-Yahoo merge risks symbol-change mismatches	data file 03	Audit permanent identifier mapping
I08	Medium	Interpretation	499 predictors include 376 interactions	Cleaning summary	Report regularization burden and run lower-dimensional sensitivity
I09	Medium	Interpretation	Unwinsorized target contains 91 observations above 100% absolute return	extreme_target_counts.csv	Report robustness to robust loss or documented winsorization
I10	Medium	Empirical credibility	No formal forecast-loss comparison test	No active test file found	Add Diebold-Mariano or panel-aware resampling in future
I11	Medium	Economic interpretation	Portfolio outputs exist but current paper focuses mostly on prediction	output/models/portfolio versus thesis	Decide whether portfolio evidence is in scope and discuss costs
I12	Medium	Reproducibility	requirements are unpinned and Python version is unspecified	requirements.txt	Lock package and Python versions
I13	Medium	Reproducibility	Input manifest stores machine-specific absolute paths	input/input_manifest.csv	Store repository-relative paths
I14	Medium	Staleness	Source caches remain for deleted scripts	orphan pyc inventory	Remove caches from submissions; do not treat them as source
I15	Low	Organization	Identical Fama-French RF files exist in input/ and input/external/	Matching SHA-1	Keep one canonical copy after confirming consumers
I16	Low	Organization	Large legacy and backup trees duplicate old model logic and outputs	backup/ and legacy/	Exclude from submission archive or label clearly

ID	Severity	Dimension	Risk	Evidence	Response
I17	Medium	Reproducibility	No automated unit-test suite is present	Repository tree	Add tests for target continuity, lags, folds, and metrics
I18	Low	Presentation	Existing thesis build auxiliaries and OS metadata are present	thesis/*.aux etc. and .DS_Store	Clean submission artifact directories
I19	Medium	Data quality	54 infinities exist before final cleaning	Dataset scan	Document affected columns and assert replacement before imputation
I20	Medium	Methodology	Beta/idio-vol formula is a rolling variance identity, not an explicit regression with intercept	data file 02:306-321	Clarify definition or compare with rolling OLS
I21	Low	Presentation	Existing bibliography contains an unresolved course-reference comment	thesis/references.bib	Verify required course citations before submission
I22	Medium	Process	Untouched-test discipline cannot be proven from code after results exist	Repository state	Describe the procedural claim cautiously

H Hard-Coded Parameter and Seed Inventory

Parameter	Value	Location	Comment
Sample	1990-01 to 2025-12	src/config.py	Price begins 1987; Jan 2026 enables final target
Train/validation end	2014-12 / 2019-12	src/config.py	Defines development/test
Daily bad-row threshold	0.10	src/config.py	Ticker-level removal
Daily/monthly extreme rules	>3 or <-0.95	data files 01/02	Data-quality flags
Risk window	252; minimum 126 daily	data file 02	Beta and idio-vol
Momentum windows	6, 12, 36 monthly observations	data file 02	Momentum excludes current return
Compustat lag	6 months	data file 03	Availability proxy
Rank interval	[-1,1]	data file 05	Monthly characteristic rank
Tuning start year	2005	tuning scripts	Actual folds end 2014
Random seed	42	active stochastic model scripts	Random Forest, Decision Tree, Elastic Net tuning
Rolling robustness	120 months	step_04_robustness	Compared with expanding
Coefficient tolerance	1e-10	step_05_test	Nonzero count

I Output Lineage

Source	Transformation	Output	Downstream use
Locked tickers	Yahoo download and quality cleaning	daily raw/clean and reports	monthly features
Daily stock + GSPC + VIX + RF	month aggregation, rolling features, shift target	monthly stock panel	Compustat merge
Raw Compustat	filter, ratios, six-month lag	clean Compustat	backward as-of panel
Fundamental panel + Welch-Goyal	SIC2 and selected variables	raw Kelly panel	imputation/ranking
Raw Kelly	monthly median, ranks, interactions	ranked parquet + predictor CSV	all active models
Fixed/tuning outputs	validation comparisons	saved parameter files	final test specifications
Ranked parquet + parameters	development fit, test prediction	test metrics/predictions	interpretation and portfolio
Frozen metrics/predictions	tables/figures/notes	interpretation directory	thesis paper

J Unused, Duplicate, Orphan, or Obsolete Inventory

Item	Evidence	Status
input/external/fama_french_rf_monthly.csv	SHA-1 identical to canonical input/fama_french_rf_monthly.csv	Duplicate, inactive
backup/model_reorganization_*	Copies of pre-reorganization model files	Archival/possibly obsolete
backup/remove_gradient_boosting_*	Removal snapshots and former portfolio scripts	Archival/possibly obsolete
src/models/legacy/gradient_boosting	Old Gradient Boosting source snapshots	Explicit legacy
output/models/legacy/gradient_boosting	Old Gradient Boosting results	Explicit legacy
orphan .pyc files	Caches exist for compare_scaling, portfolio, old robustness/test, and inspection sources that are absent	Temporary/stale
wg_infl	Present in Welch-Goyal input but absent from locked macro columns	Unused input column
notebooks/	Empty directory	Nonessential
Thesis build auxiliaries	Standard LaTeX auxiliary files	Regenerable temporary files

K Final Submission Checklist

Priority	Issue	File	How to verify	Change required	Difficulty
A - must	Target continuity	data file 02 and monthly panel	Confirm or rebuild only when next month is exactly $t+1$	Code change required	Moderate
A - must	Macro lag	data file 04 and Welch-Goyal source timing	Verify release availability and apply documented lag	Likely code change	Moderate
A - must	Canonical run order	run_pipeline.sh	Replace missing paths after deciding portfolio scope	Code/doc change	Easy
A - must	Tuning description	models README and estimation code	Choose actual minimum-MSE train-only description or revise method	Documentation or code	Easy
A - must	Course references	thesis/references.bib	Verify professor-required citations	Documentation only	Easy
B - should	Yahoo failures	data file 01	Recover failed list from logs or redownload metadata only	Potential code change	Moderate
B - should	Ticker/Compustat mapping	input and merged panel	Sample unmatched/symbol-change cases	No immediate code change	Difficult
B - should	Environment lock	requirements.txt	Record tested versions and platform	Config change	Easy
B - should	Portfolio scope	output/models/portfolio and paper	Decide inclusion; disclose no costs	Documentation only	Easy
C - future	Point-in-time universe	universe/price data	Use CRSP identifiers and delisting returns	Major rebuild	Difficult
C - future	Formal forecast tests	prediction output	Add time/panel-aware loss comparison	New analysis	Moderate
C - future	Robust target/loss	cleaning and modeling	Compare clipping, Huber loss, or robust evaluation	New analysis	Moderate
D - optional	Alternative models	model stages	Add only under prespecified chronological validation	New analysis	Difficult

L Assumption and Readiness Register

1. Yahoo adjusted prices are treated as usable but are not independently validated against CRSP.
2. The supplied raw Compustat file is assumed to be the intended extraction; query provenance is absent.
3. A six-month reporting lag is a conservative proxy, not an actual filing date.
4. Same-month market variables are assumed observable at month-end for next-month prediction.
5. Welch-Goyal availability timing is unresolved.
6. The 2020-2025 test is treated as final, but viewing history is not verifiable.
7. Frozen outputs are treated as the current run because timestamps and internal values align; source-to-output hashes are absent.

Readiness conclusion: Requires important checks. The report identifies 22 evidence-supported issues, of which four are high severity.